

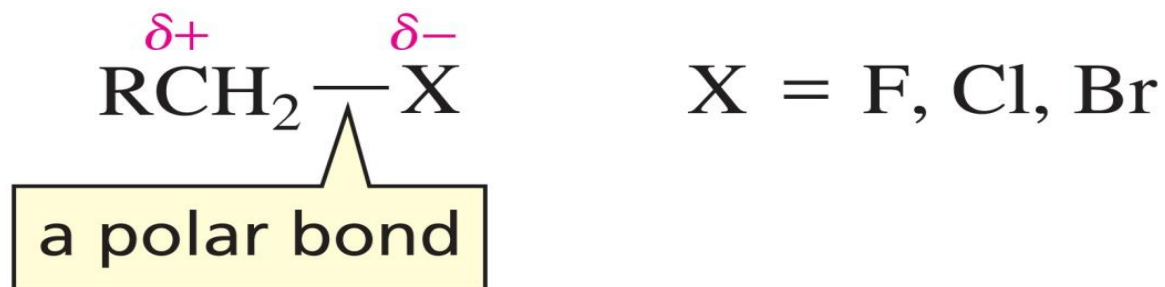
Chapter 8

SUBSTITUTION AND ELIMINATION REACTIONS OF ALKYL HALIDES

Alkyl Halides are hydrocarbons that contain the C-X bond somewhere in the molecule



R-X bond is polar bond due to electronegativity difference between C and the halogen



Copyright © 2007 Pearson Prentice Hall, Inc.

The reactivity of RX depends on X

What is a substitution reaction?

S_N2 or S_N1

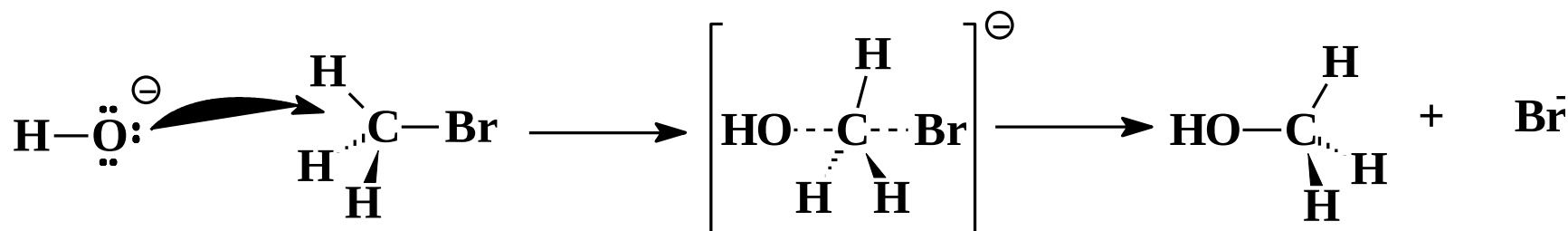
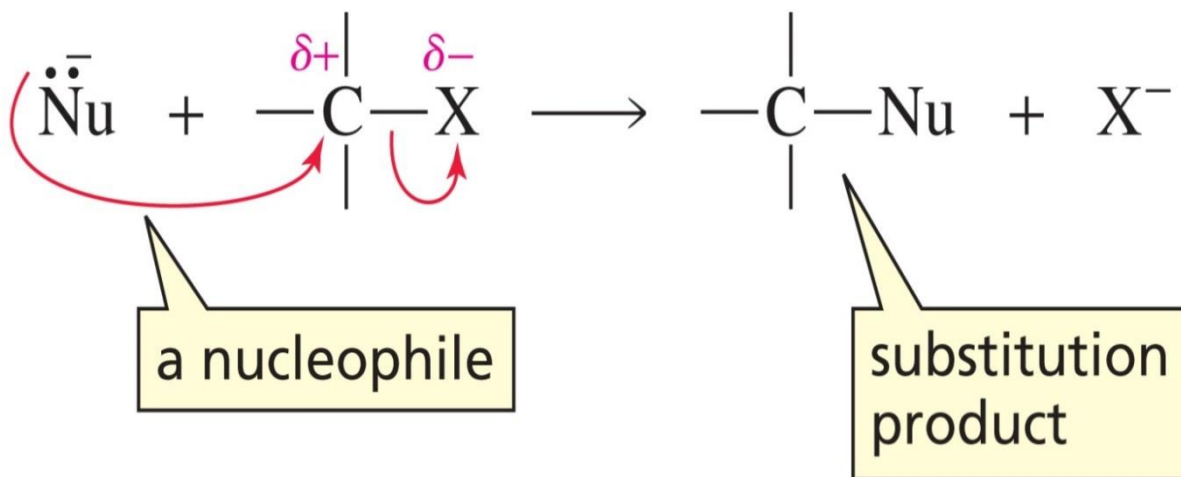


$E2$ or $E1$

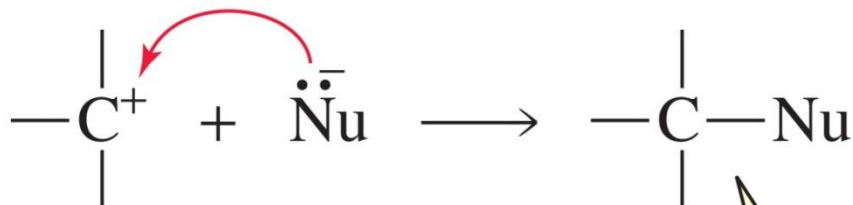
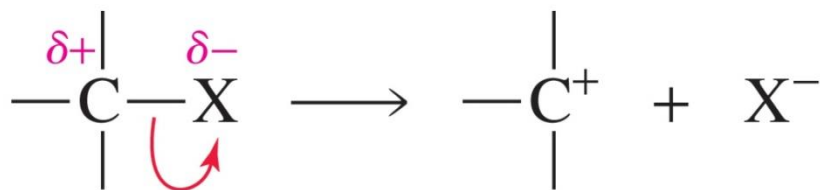
The atom or group that is substituted or eliminated in these reactions is called a leaving group.

How do alkyl halides react?

S_N2



S_N1



substitution
product



The substitution is more precisely called a **nucleophilic substitution** because the atom or group replacing the leaving group is a **nucleophile**

The reaction mechanism which predominates depends on the following factors:

1. The structure of the alkyl halide
2. The concentration & reactivity of the nucleophile
3. The nature of the leaving group
4. The solvent of the reaction

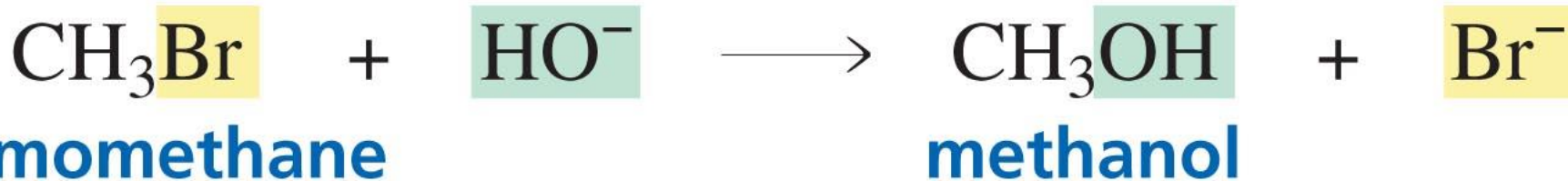
Two types of Nucleophilic(N) Substitution(S)

1. Bimolecular (2 molecules): S_N2
2. Unimolecular (1 Molecule): S_N1

Experimental Evidence for the S_N2 Reaction Mechanism

1. The rate of the reaction is dependent on the concentration of both the alkyl halide and the nucleophile **second order reaction kinetics**
2. The rate of the reaction with a given nucleophile decreases with increasing branching of the alkyl halide at the reacting center.
3. The configuration of the substituted product is inverted compared to the configuration of the reacting chiral alkyl halide.

The Rate Law of an S_N2 Reaction



© 2011 Pearson Education, Inc.

Obtained experimentally:

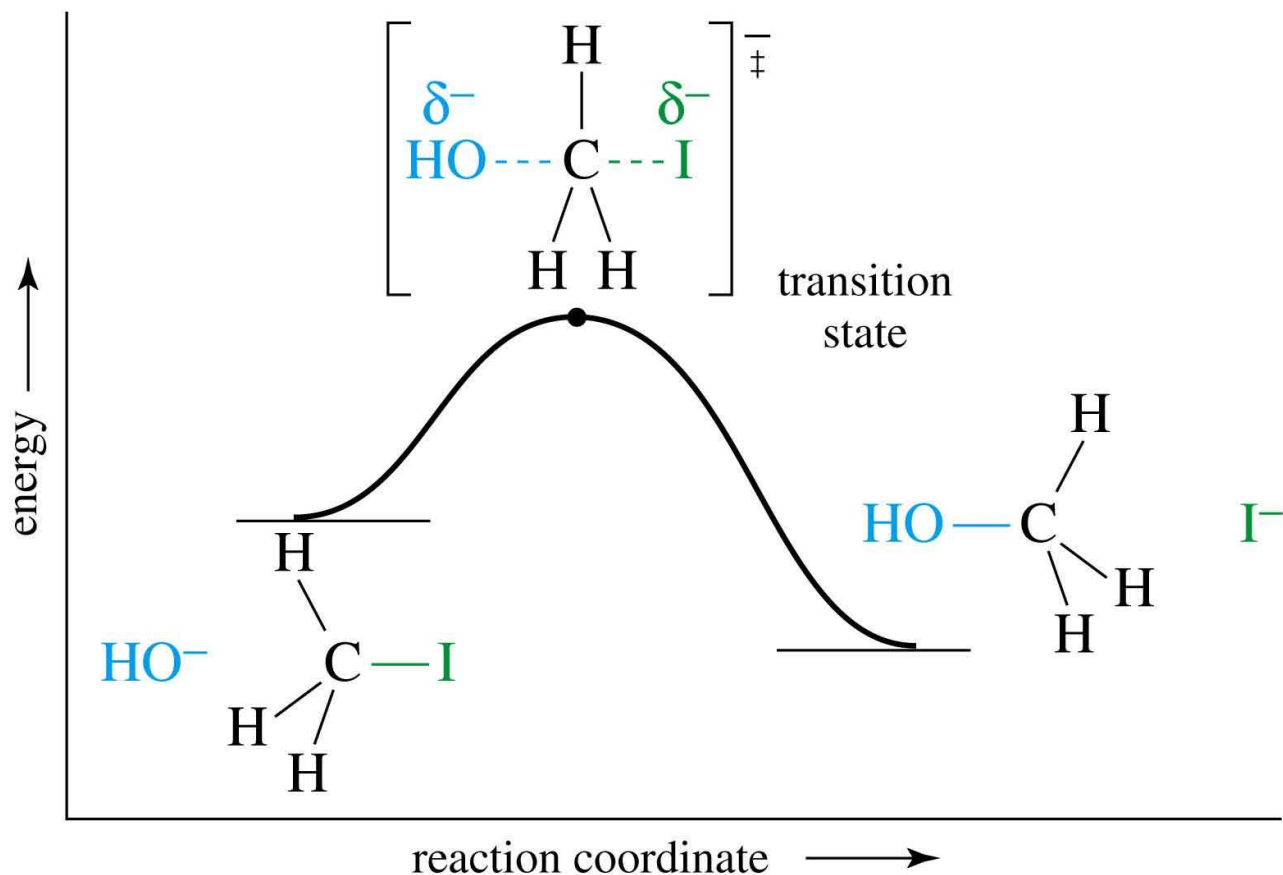
$$\text{rate} = k [\text{alkyl halide}][\text{nucleophile}]$$

the rate constant

Rate law includes both the alkyl halide and the nucleophile, a second-order process

© 2011 Pearson Education, Inc.

S_N2 Energy Diagram



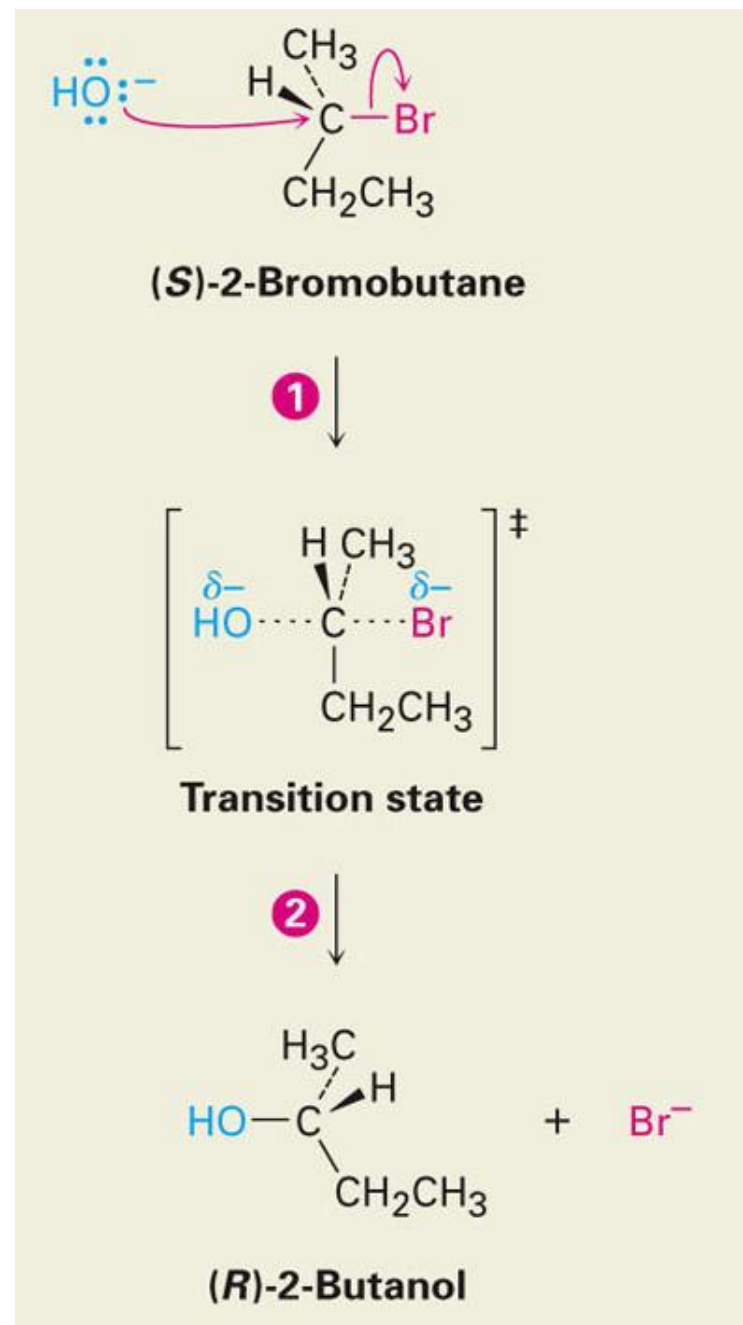
- One-step reaction

Inversion of chiral center (**Walden inversion**)

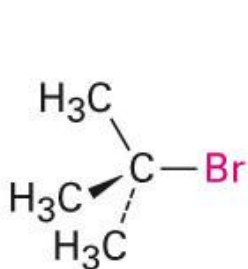
Sensitive to steric effects

Methyl halides are most reactive ($\text{CH}_3\text{-X}$)

$\text{CH}_3\text{-X} > \text{primary} > \text{secondary} > \text{tertiary}$

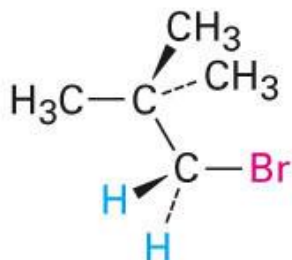


The Influence of Branching on the S_N2 Rate



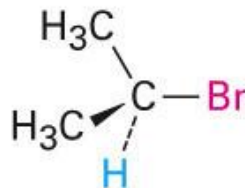
Tertiary

< 1



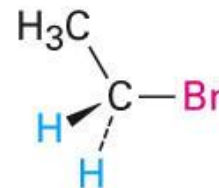
Neopentyl

1



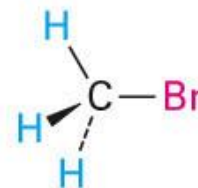
Secondary

500



Primary

40,000



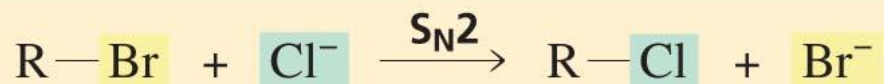
Methyl

2,000,000

Relative
reactivity



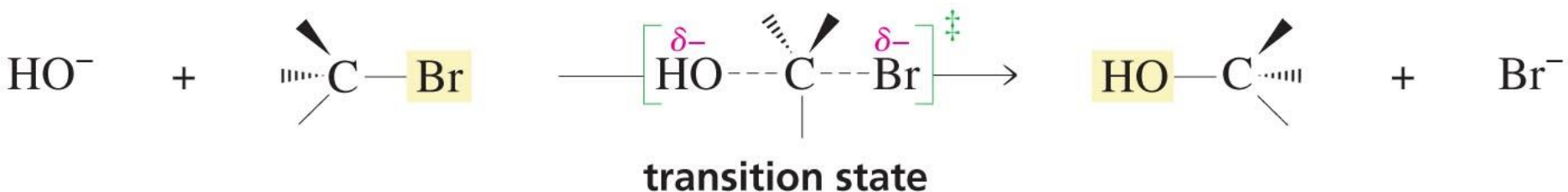
Table 8.1 Relative Rates of S_N2 Reactions for Several Alkyl Halides



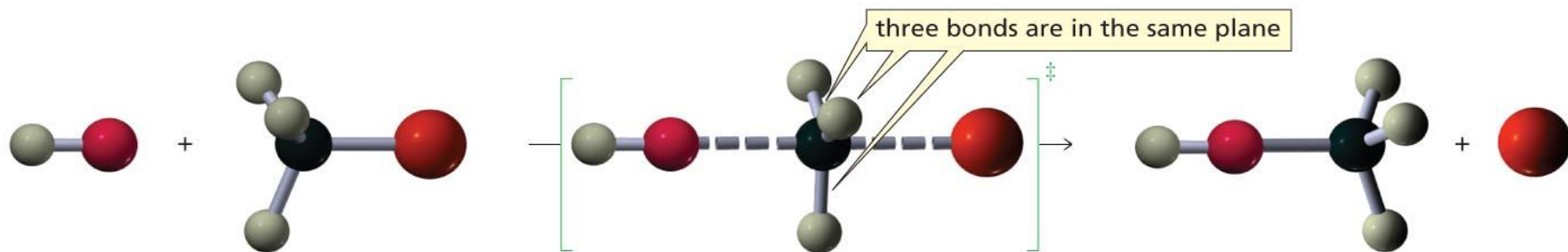
Alkyl halide	Class of alkyl halide	Relative rate
CH ₃ —Br	methyl	1200
CH ₃ CH ₂ —Br	primary	40
CH ₃ CH ₂ CH ₂ —Br	primary	16
CH ₃ CH—Br	secondary	1
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 \\ \\ \text{CH}_3\text{C}-\text{Br} \\ \\ \text{CH}_3 \end{array}$	tertiary	too slow to measure

Why Does Branching Lower the S_N2 Rate?

Because the nucleophile attacks the back side of the carbon that is bonded to the leaving group:

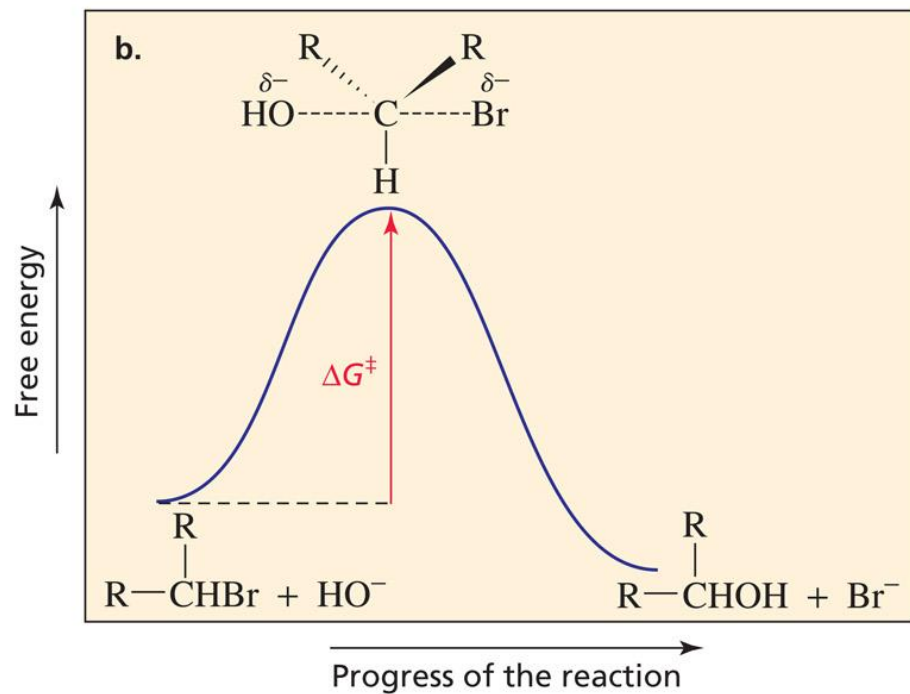
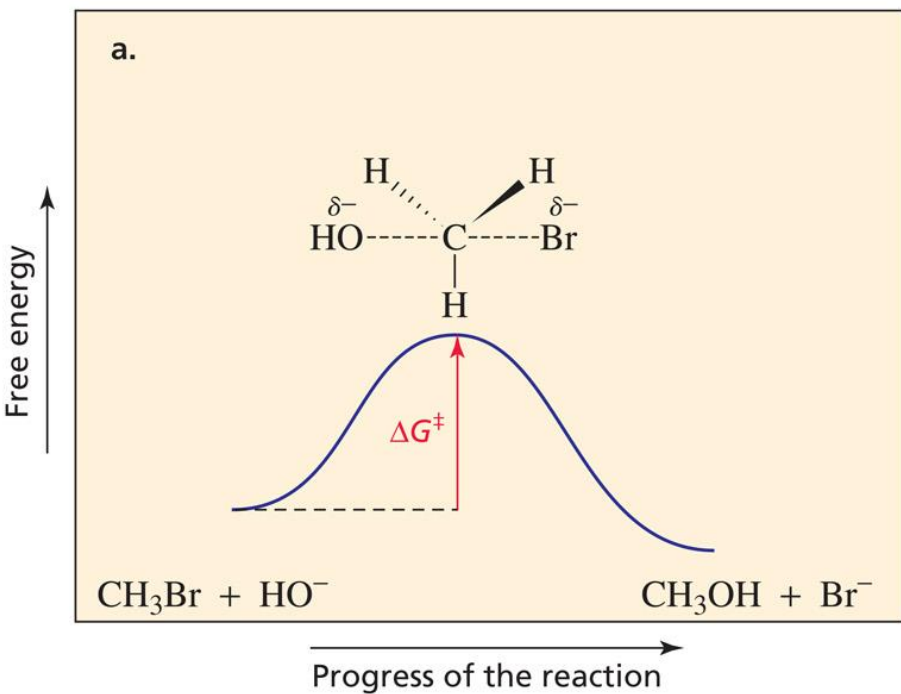


© 2011 Pearson Education, Inc.



© 2011 Pearson Education, Inc.

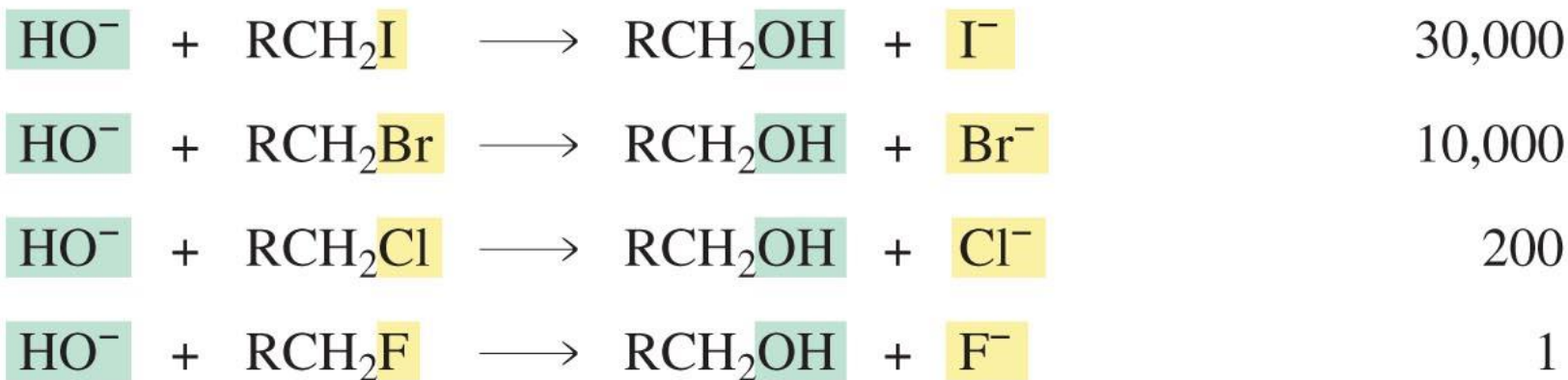
Tertiary alkyl halides cannot undergo S_N2 reactions



Copyright © 2006 Pearson Prentice Hall, Inc.

S_N2 Reactions Are Affected by the Leaving Group

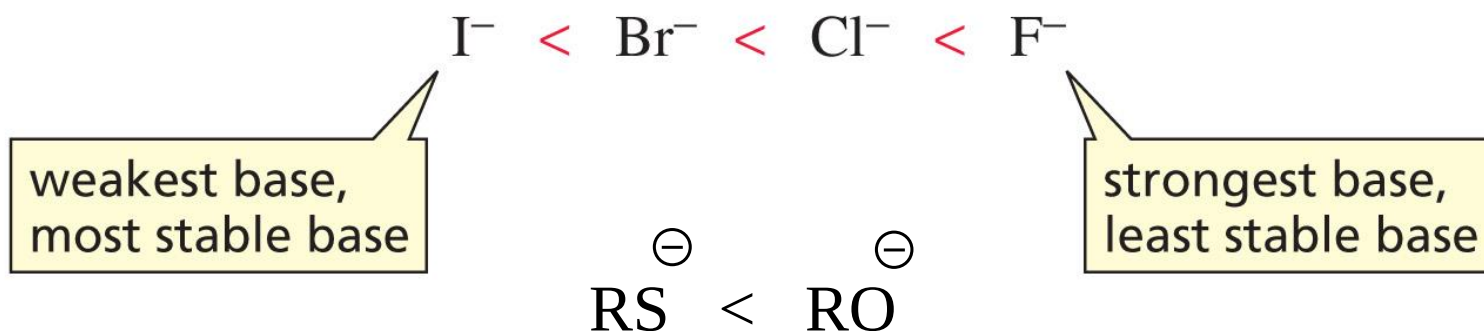
relative rates of reaction



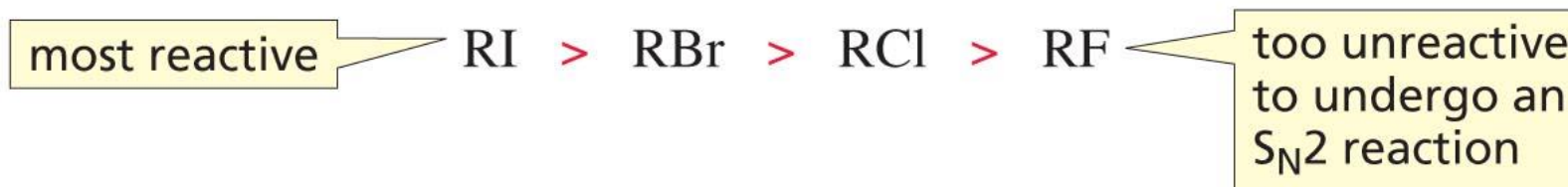
© 2011 Pearson Education, Inc.

Good leaving groups are the conjugate bases of strong acids, i.e., they are weak bases:

relative basicities of the halide ions



relative reactivities of alkyl halides in an $\text{S}_{\text{N}}2$ reaction



Nucleophiles in the S_N2 Reaction

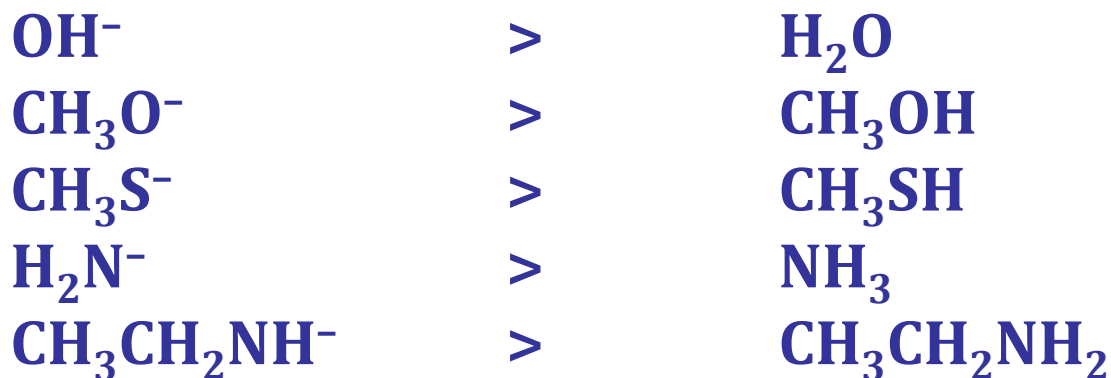
Second-row nucleophiles:

- Are approximately the same size.
- The stronger the base, the better the nucleophile.

conjugate acid-base pair, the base is stronger

stronger base,
better nucleophile

weaker base,
poorer nucleophile



Therefore the strength of second-row nucleophiles is determined by conjugate acid strength:

Decreases left to right on Periodic Table. More electronegative atoms less likely to form new bond:

relative acid strengths

weakest
acid



© 2011 Pearson Education, Inc.

relative base strengths and relative nucleophilicities

strongest
base



best nucleophile

© 2011 Pearson Education, Inc.

Solvent effect

The solvent can be classified into:

1- nonpolar Solvent

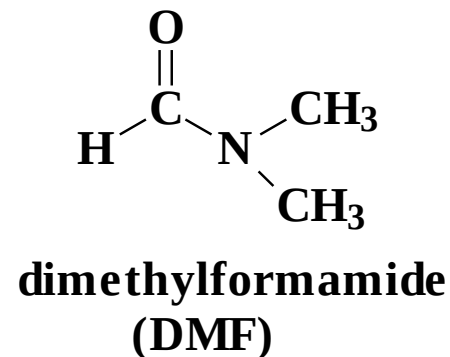
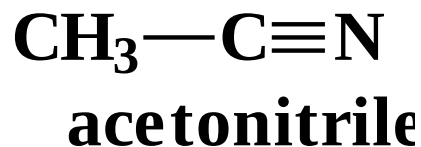
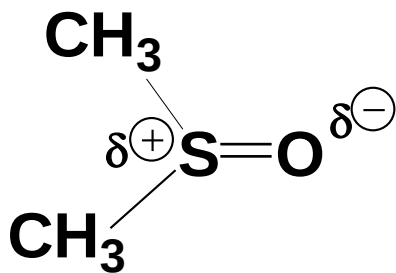
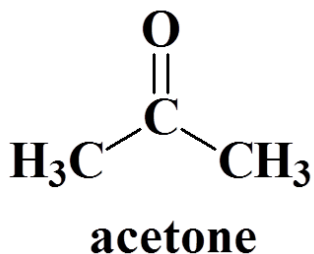
2- Polar Solvent

a) Polar protic solvent

can form hydrogen bond (H_2O , amines....etc)

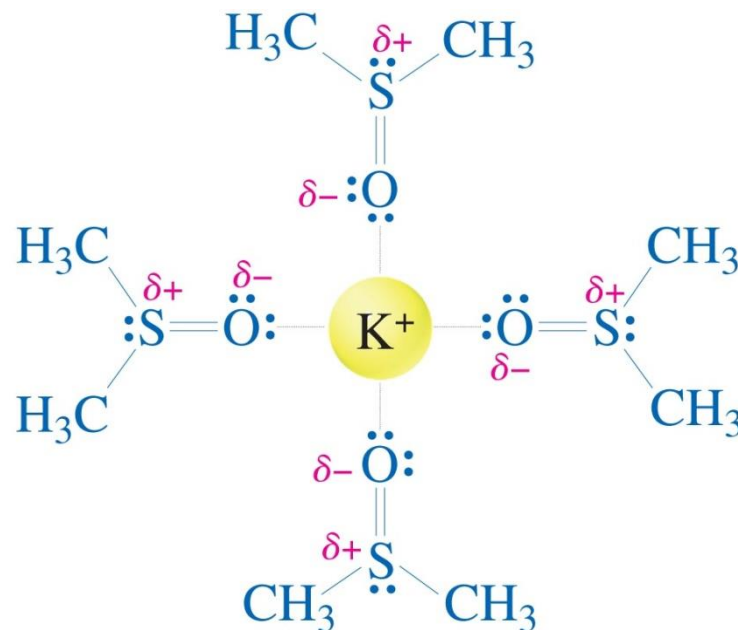
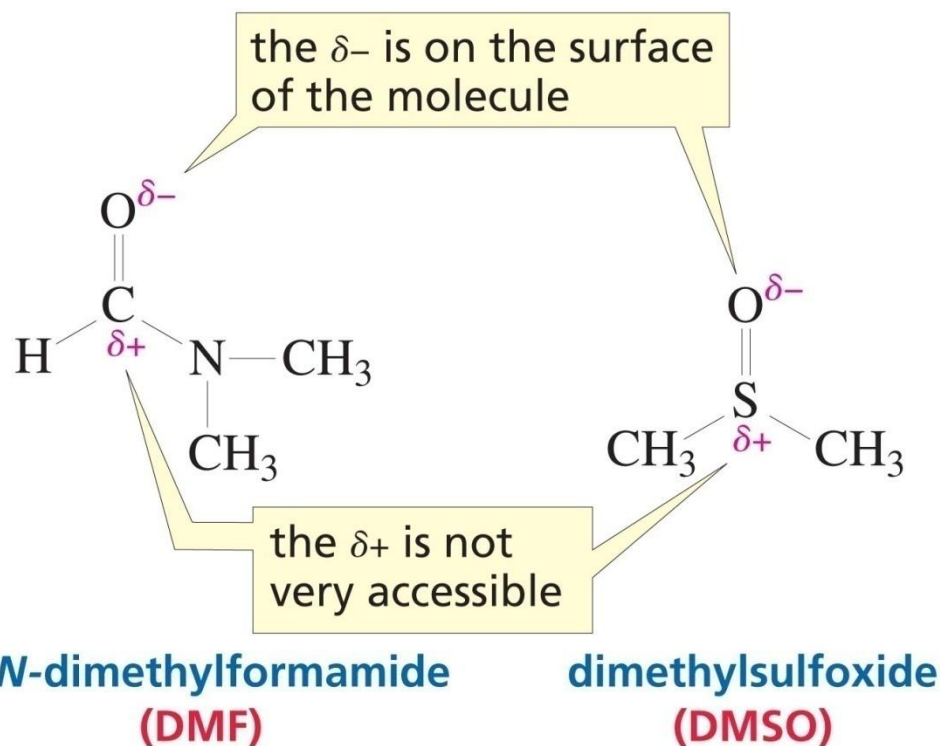
b) Polar aprotic solvent

can't form hydrogen bond (acetone, DMSO,....)



S_N2 reactions are carried out in polar aprotic solvents:

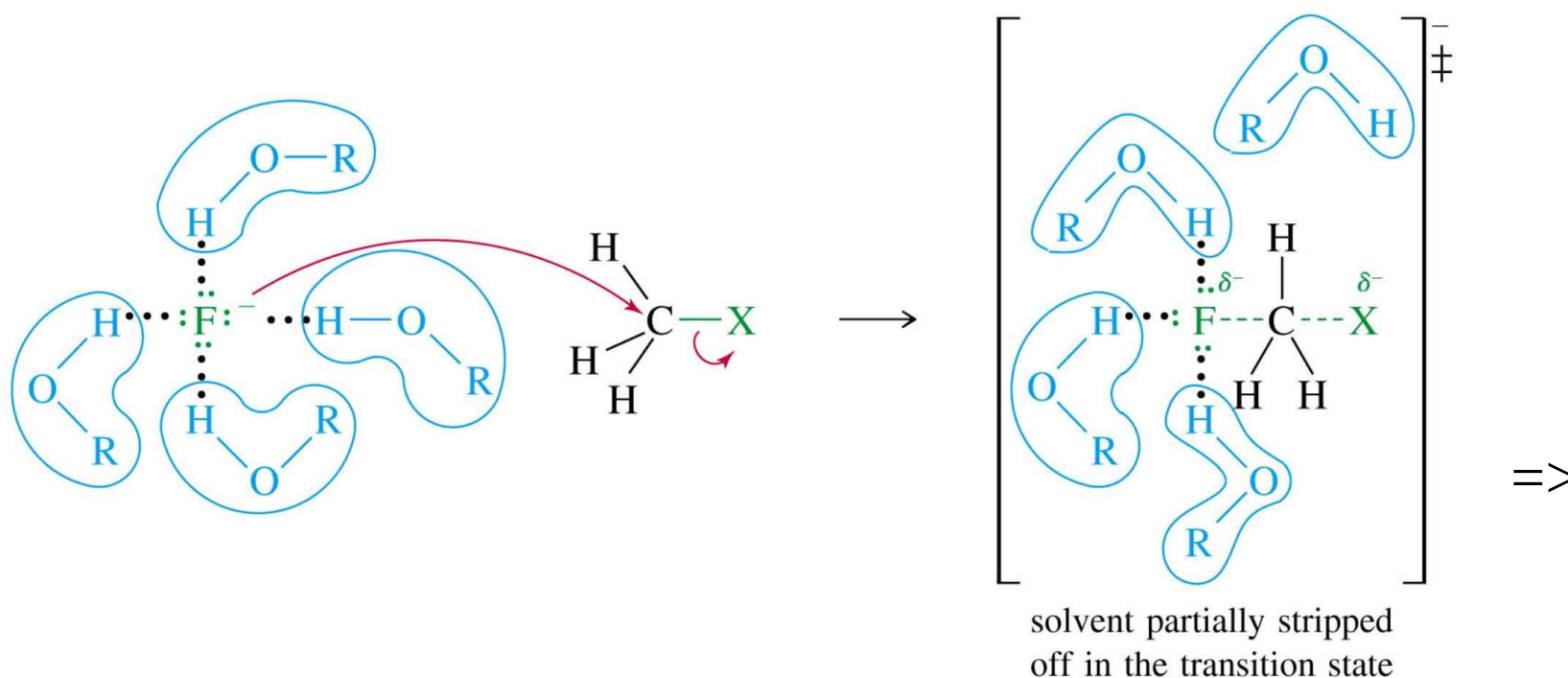
Includes **DMSO**, **DMF**, **acetone** and **acetonitrile** (CH₃CN)



DMSO can solvate a cation better than it can solvate an anion

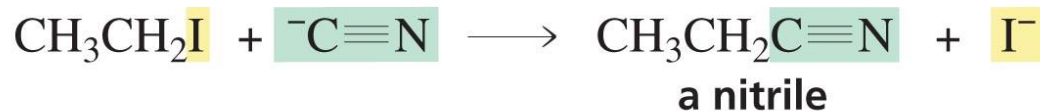
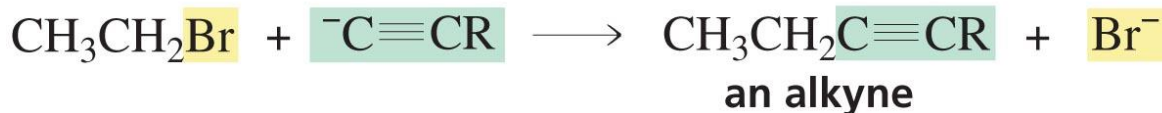
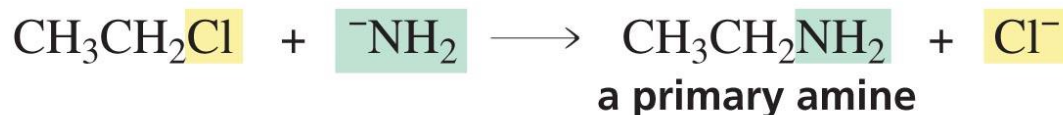
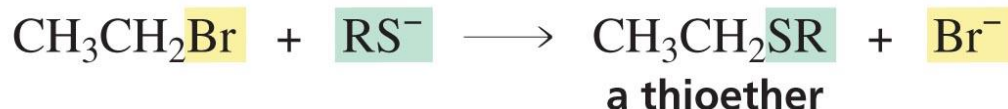
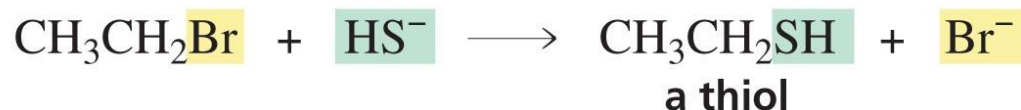
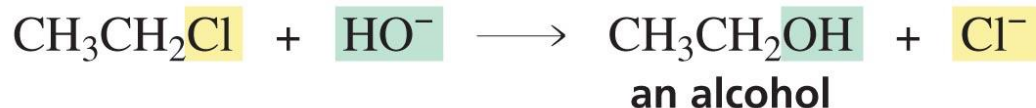
Consequently, the anionic nucleophile is unsolvated and reactive

Polar protic solvents (O-H or N-H) reduce the strength of the nucleophile. Hydrogen bonds must be broken before nucleophile can attack the carbon.



Synthetic Utility of the S_N2 Reaction

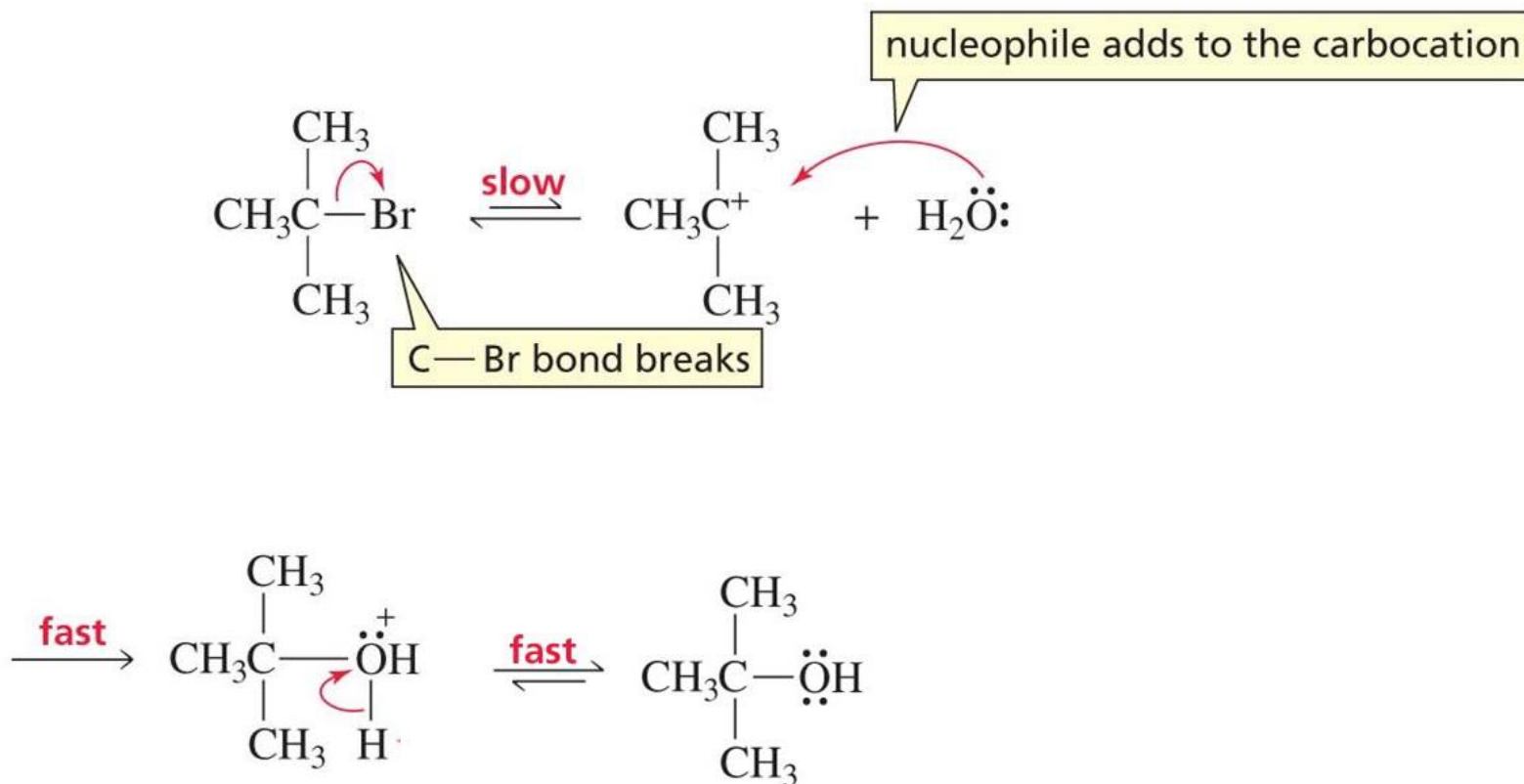
A variety of functional groups can be prepared employing a good nucleophile and an electrophile with a good leaving group:



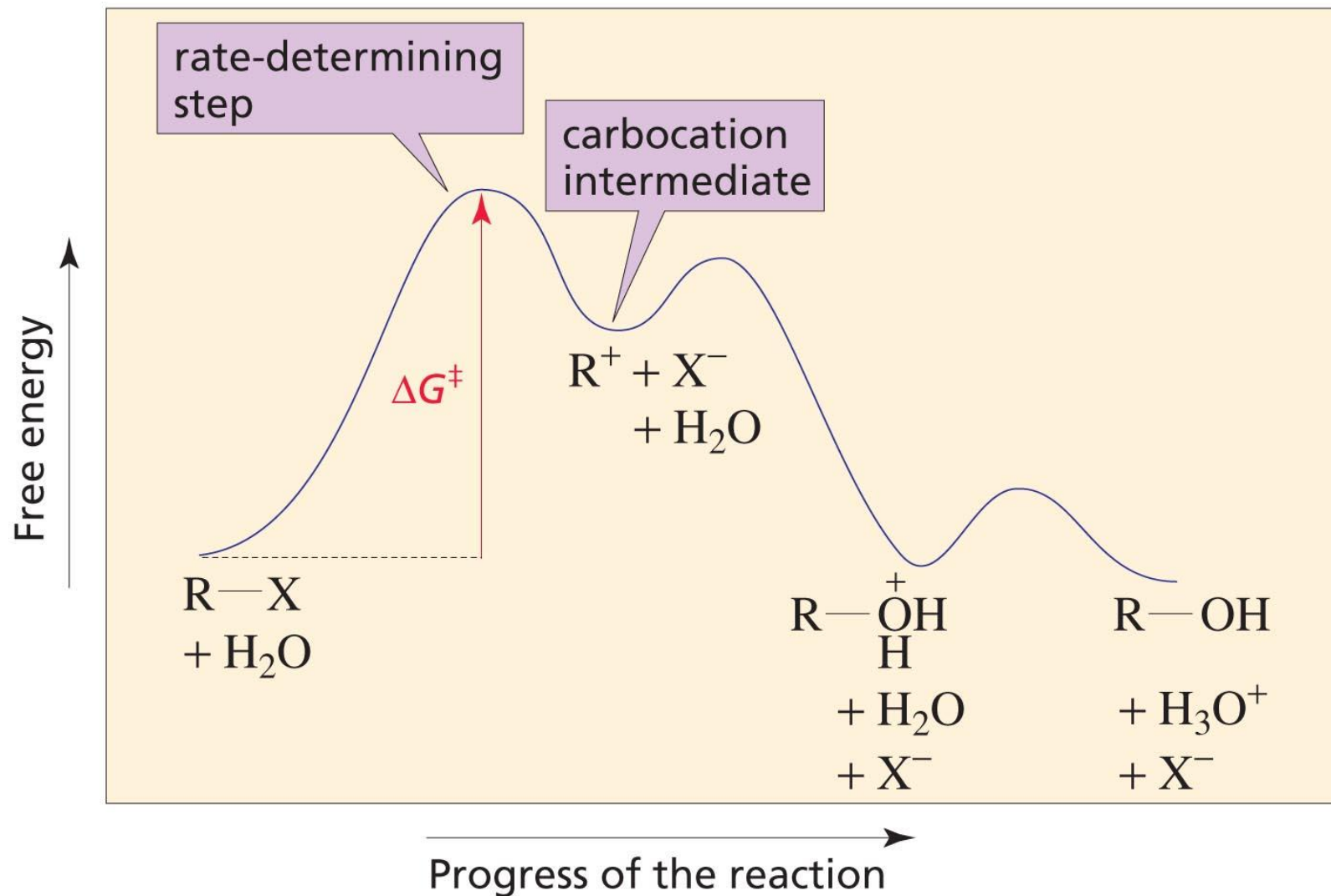
Experimental Evidence for an S_N1 Reaction

1. The rate of the reaction depends only on the concentration of the alkyl halide.
2. The rate of the reaction increases with branching of the alkyl halide at the reacting center.
3. An S_N1 reaction with an enantiomeric pure alkyl halide affords a racemic mixture

An **S_N1** is a two-step reaction and the leaving group departs before the nucleophile approaches:



Reaction Coordinate Diagram for an S_N1 Reaction



Influence of Alkyl Halide Branching at the Reacting Center

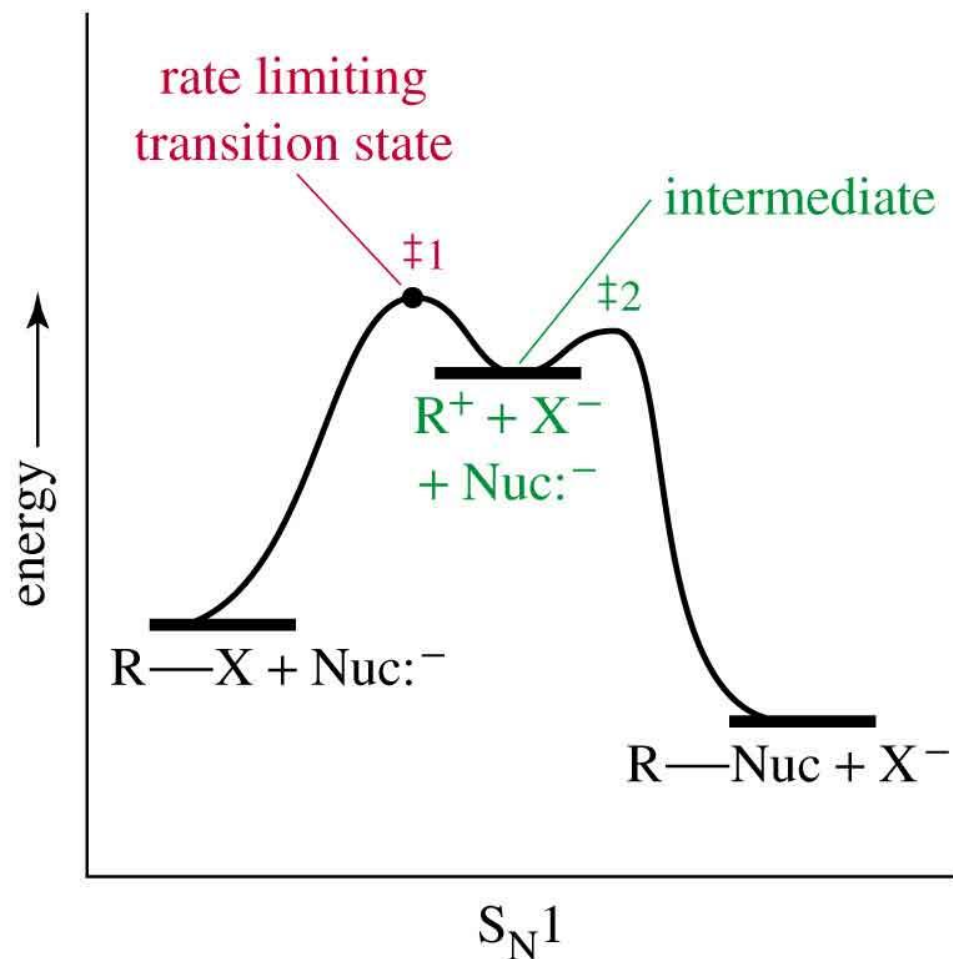
Table 8.4 Relative Rates of S_N1 Reactions for Several Alkyl Bromides (solvent is H_2O , nucleophile is H_2O)

Alkyl bromide	Class of alkyl bromide	Relative rate
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3\text{C}-\text{Br} \\ \\ \text{CH}_3 \end{array}$	tertiary	1,200,000
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3\text{CH}-\text{Br} \\ \\ \text{CH}_3 \end{array}$	secondary	11.6
$\text{CH}_3\text{CH}_2-\text{Br}$	primary	1.00*
CH_3-Br	methyl	1.05*

*Although the rate of the S_N1 reaction of this compound with water is 0, a small rate is observed as a result of an S_N2 reaction.

S_N1 Energy Diagram

- Forming the carbocation is endothermic
- Carbocation intermediate is in an energy well.

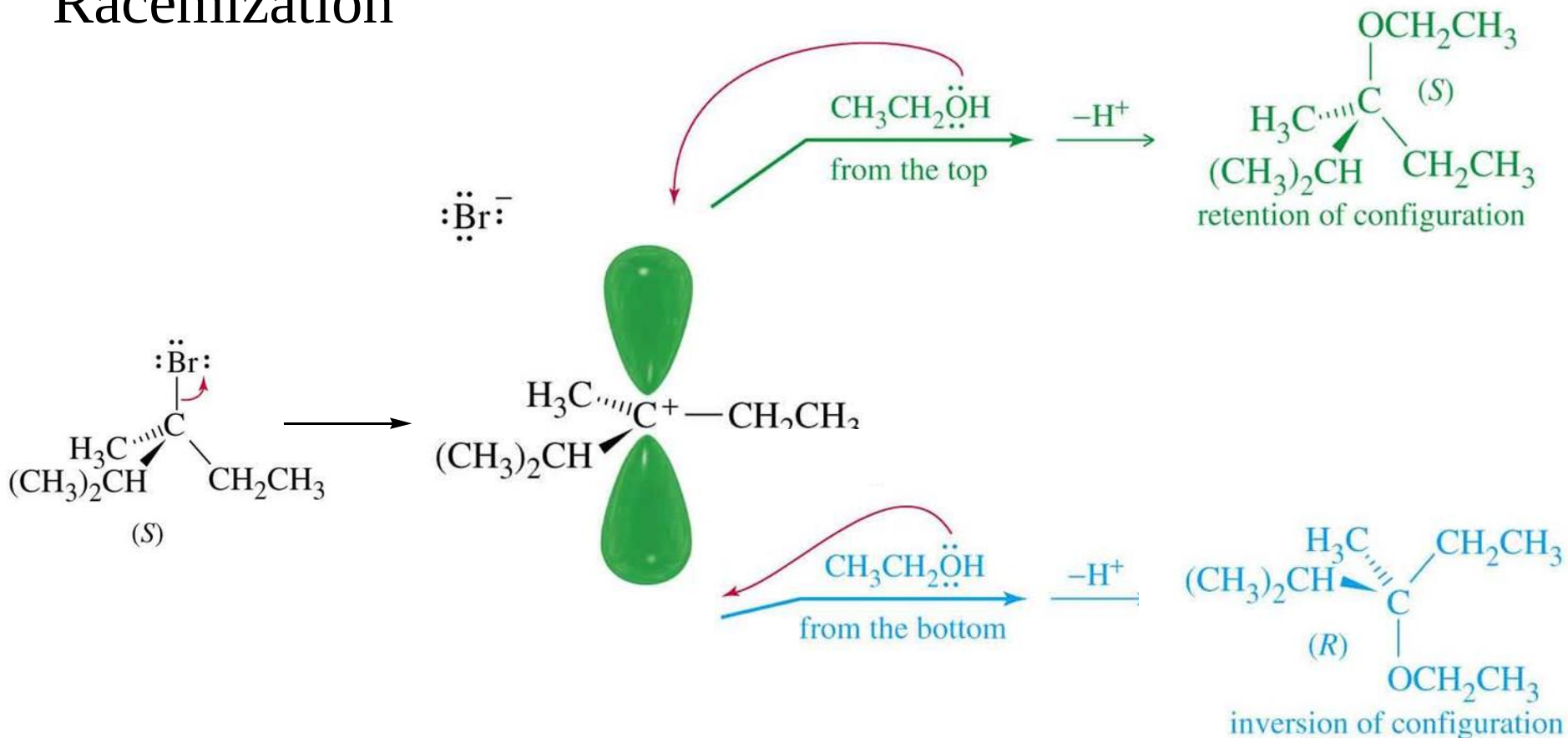


Rates of S_N1 Reactions

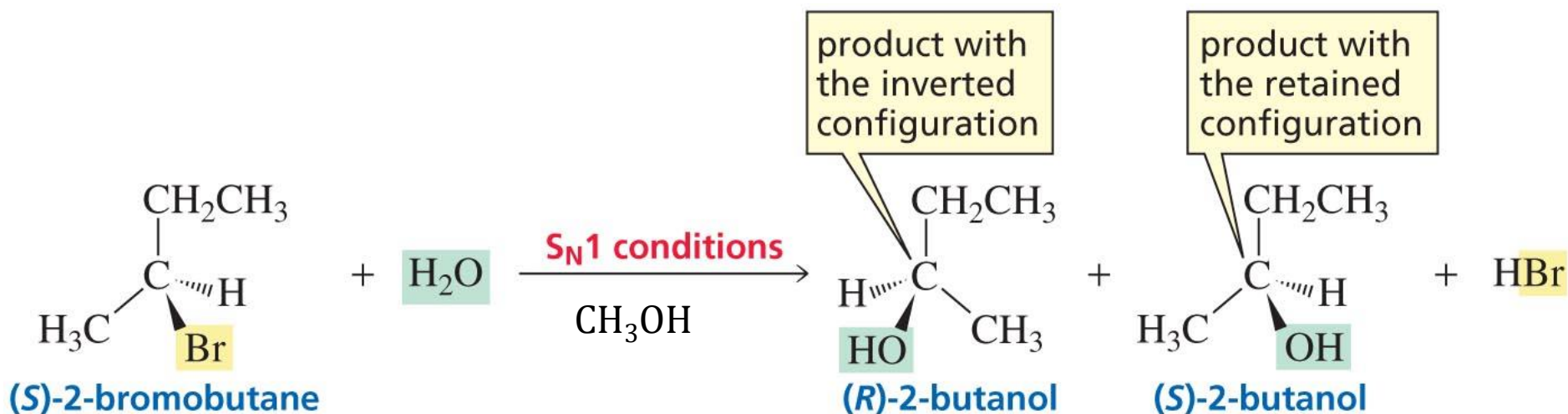
- $3^\circ > 2^\circ > 1^\circ \gg \text{CH}_3\text{X}$
 - Order follows stability of carbocations (opposite to S_N2)
 - More stable ion requires less energy to form
- Better leaving group, faster reaction (like S_N2)
- Polar protic solvent best: It solvates ions strongly with hydrogen bonding.
- The higher the polarity of the solvent, the larger the rate.
- The nucleophile concentration has no effect on the rate of an S_N1 reaction because it is not in the rate-determining step.

Stereochemistry of S_N1

Racemization



The S_N1 reaction of an enantiomeric pure alkyl halide affords a racemic mixture:



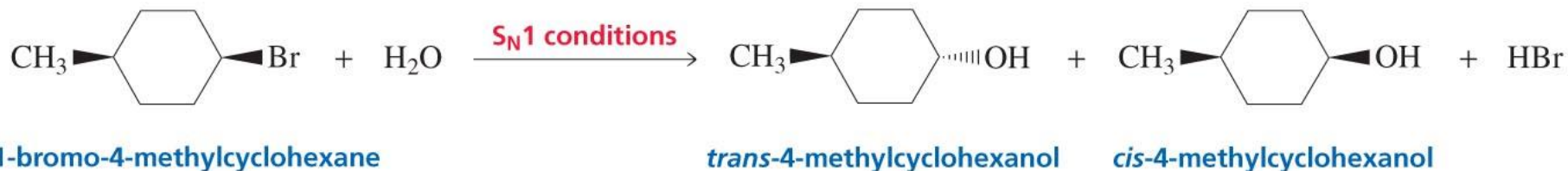
© 2011 Pearson Education, Inc.

The products resulting from substitution of cyclic compounds:

Inversion versus racemization



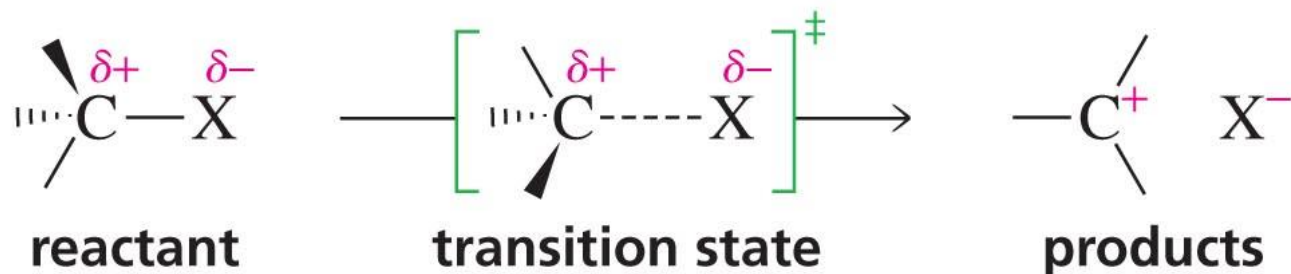
© 2011 Pearson Education, Inc.



© 2011 Pearson Education, Inc.

The effect of a solvent on the rate of an S_N1 reaction:

rate-determining step of an S_N1 reaction



© 2011 Pearson Education, Inc.

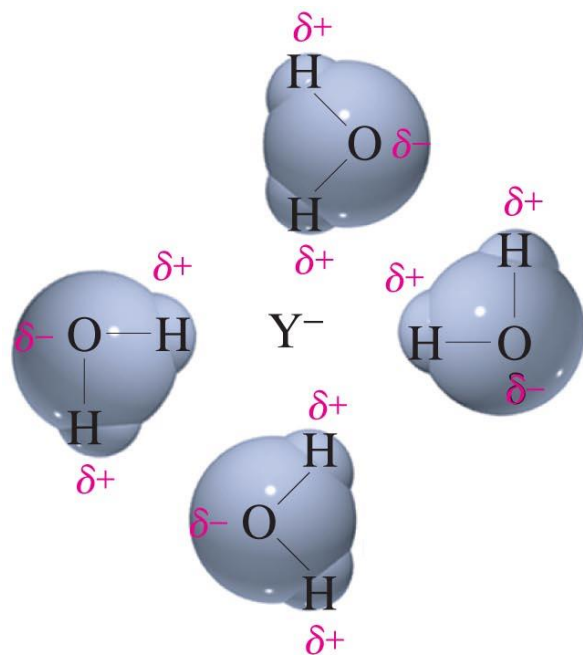
Table 8.8 The Effect of the Polarity of the Solvent on the Rate of Reaction of 2-Bromo-2-methylpropane in an S _N 1 Reaction	
Solvent	Relative rate
100% water	1200
80% water / 20% ethanol	400
50% water / 50% ethanol	60
20% water / 80% ethanol	10
100% ethanol	1

© 2011 Pearson Education, Inc.

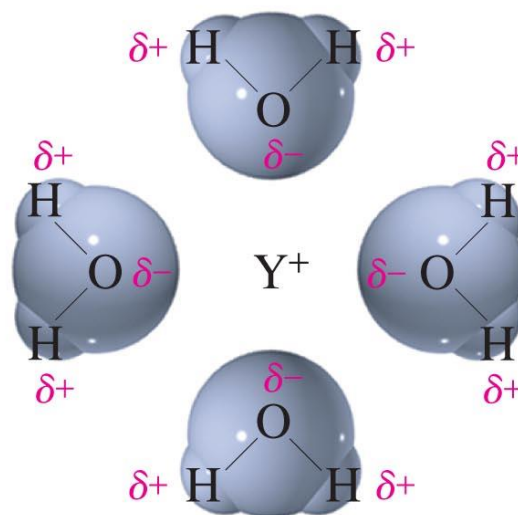
The dielectric constant is a measure of how the solvent can insulate opposite charges from one another:

H_2O (ϵ 79) is a polar, protic solvent.

$\text{CH}_3\text{CH}_2\text{OH}$ (ethanol, ϵ 25) is also protic, but much less polar than H_2O .

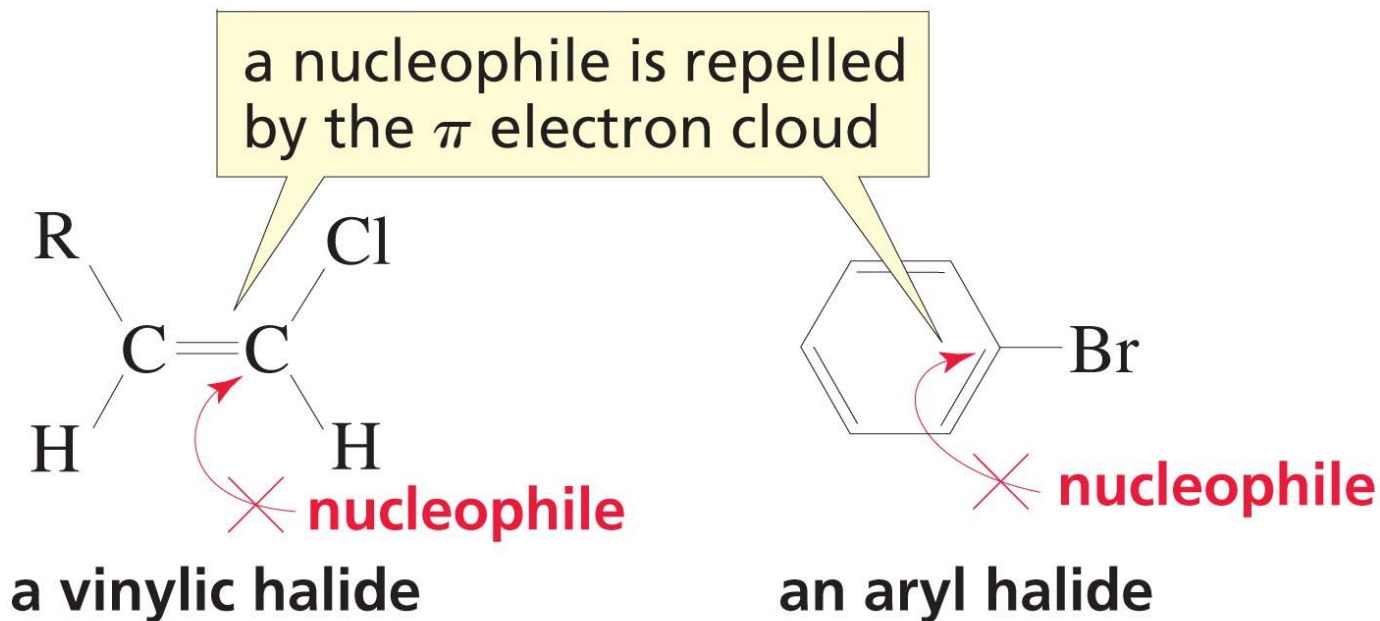


ion-dipole interactions
between a negatively
charged species and water



ion-dipole interactions
between a positively
charged species and water

Vinyl and aryl halides **do not undergo** S_N2 because:



© 2011 Pearson Education, Inc.

Vinyl and aryl halides **do not undergo** S_N1 because:

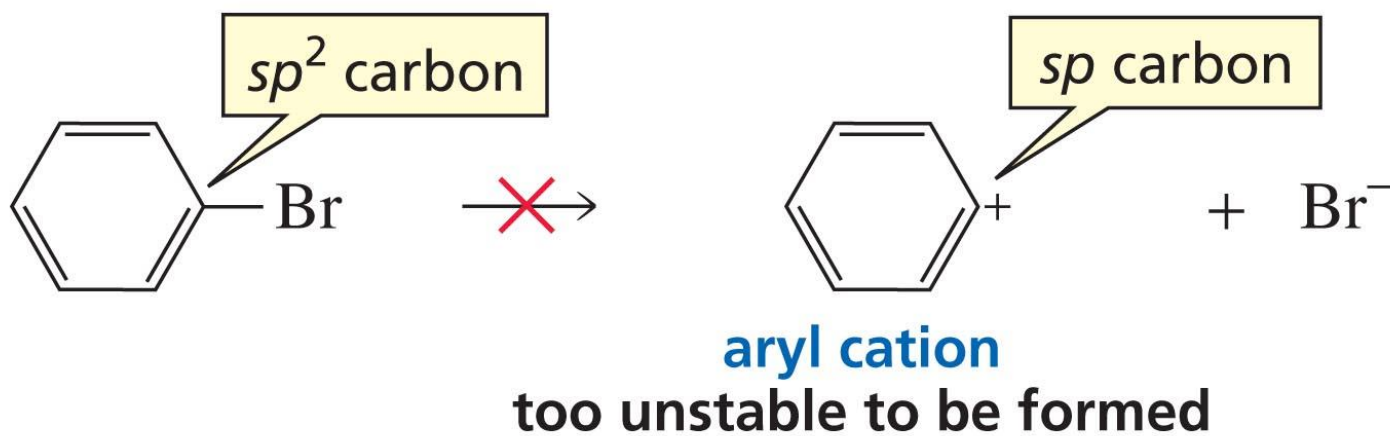
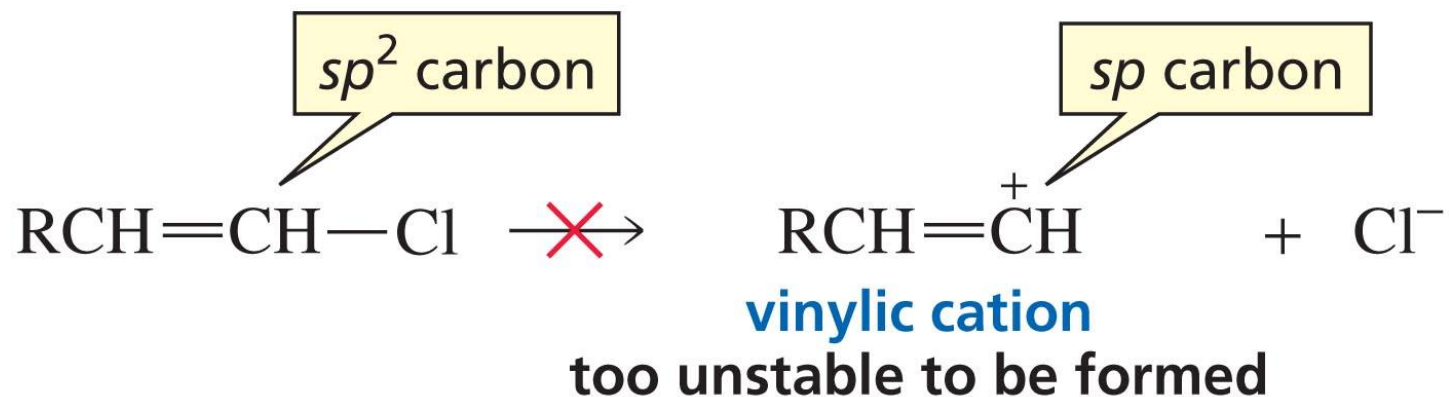


Table 8.5 Comparison of S_N2 and S_N1 Reactions

S _N 2	S _N 1
A one-step mechanism	A stepwise mechanism with a carbocation intermediate
A bimolecular rate-determining step	A unimolecular rate-determining step
No carbocation rearrangements	Carbocation rearrangements
The better the nucleophile, the faster the rate of the reaction.	The strength of the nucleophile does not affect the rate of the reaction.
Reactivity order: methyl > 1° > 2° > 3° (No reaction with tertiary.)	Reactivity order: 3° > 2° > 1° > methyl (No reaction with primary or methyl.)
Product has the inverted configuration relative to that of the reactant.	Products have both the retained and the inverted configurations relative to that of the reactant.
Leaving group: I ⁻ > Br ⁻ > Cl ⁻ > F ⁻	Leaving group: I ⁻ > Br ⁻ > Cl ⁻ > F ⁻

© 2011 Pearson Education, Inc.

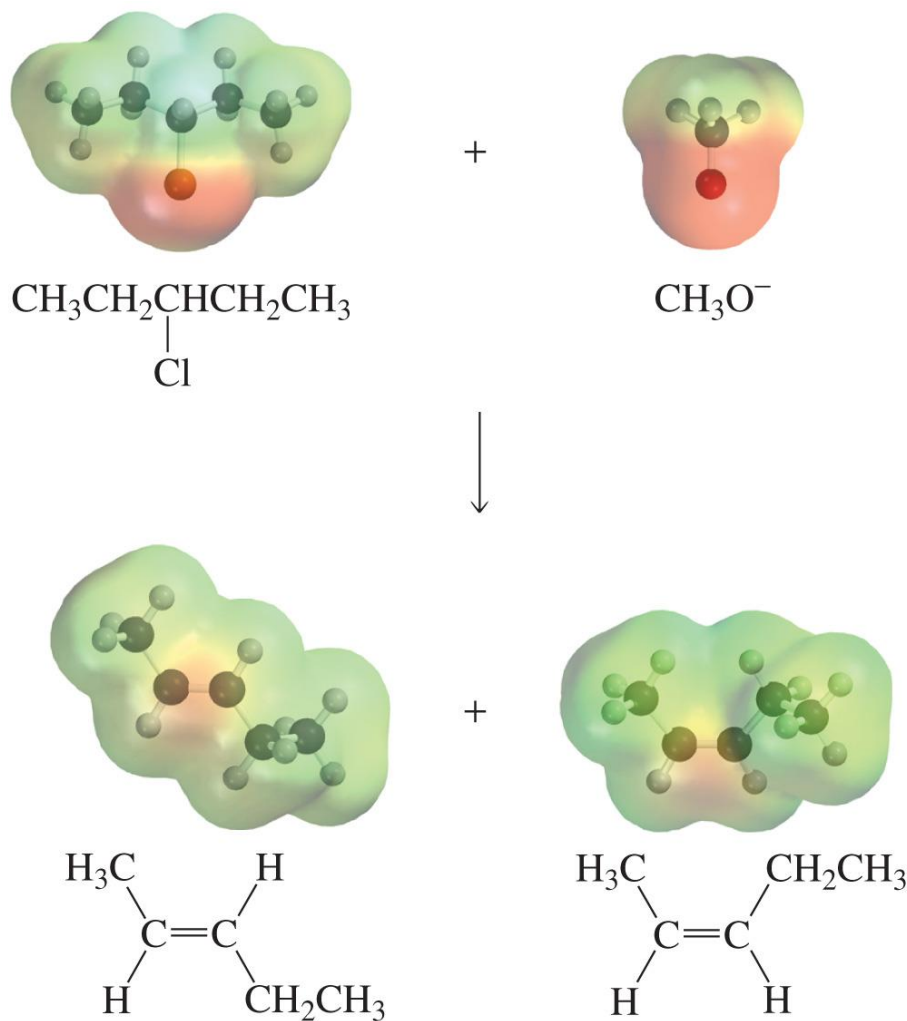
Table 8.6 Summary of the Reactivity of Alkyl Halides in Nucleophilic Substitution Reactions

Methyl and 1° alkyl halides	S _N 2 only
Vinylic and aryl halides	Neither S _N 1 nor S _N 2
2° alkyl halides	S _N 1 and S _N 2
1° and 2° benzylic and 1° and 2° allylic halides	S _N 1 and S _N 2
3° alkyl halides	S _N 1 only
3° benzylic and 3° allylic halides	S _N 1 only

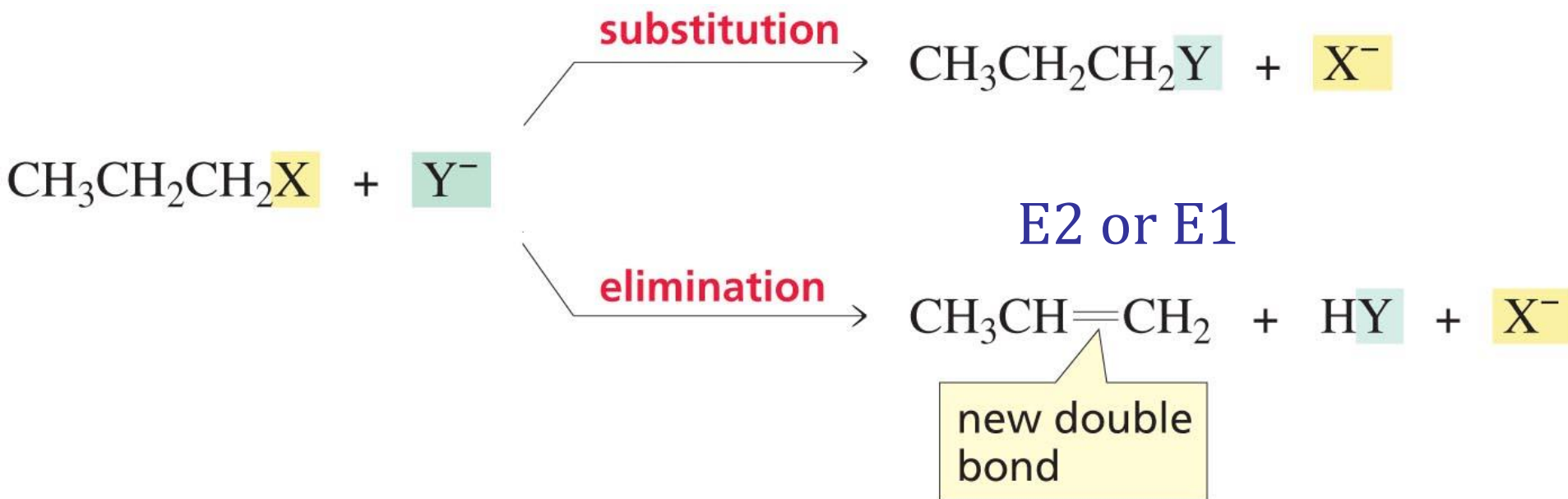
© 2011 Pearson Education, Inc.

Elimination Reactions of Alkyl Halides

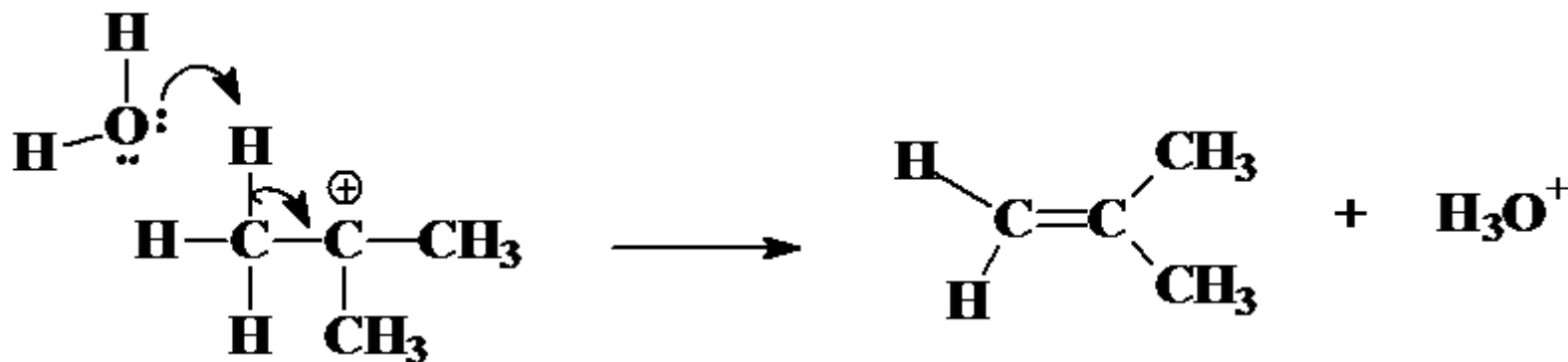
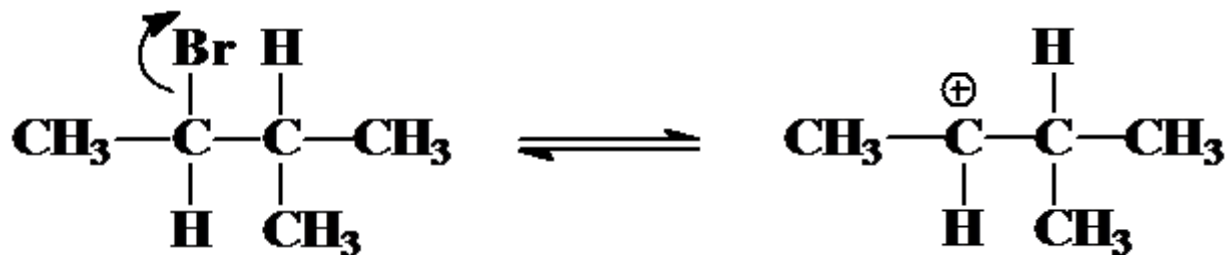
Competition Between Substitution and Elimination

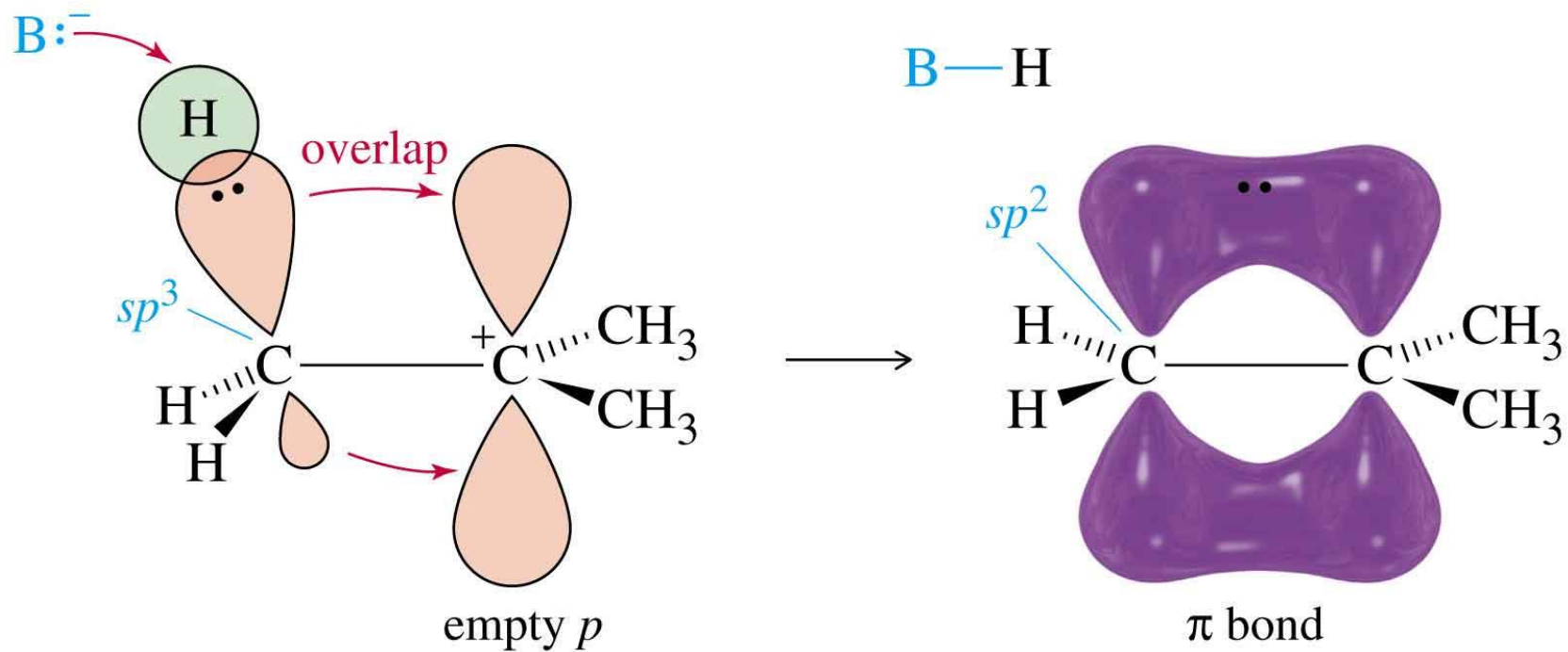


In addition to substitution, an alkyl halide can undergo an elimination reaction

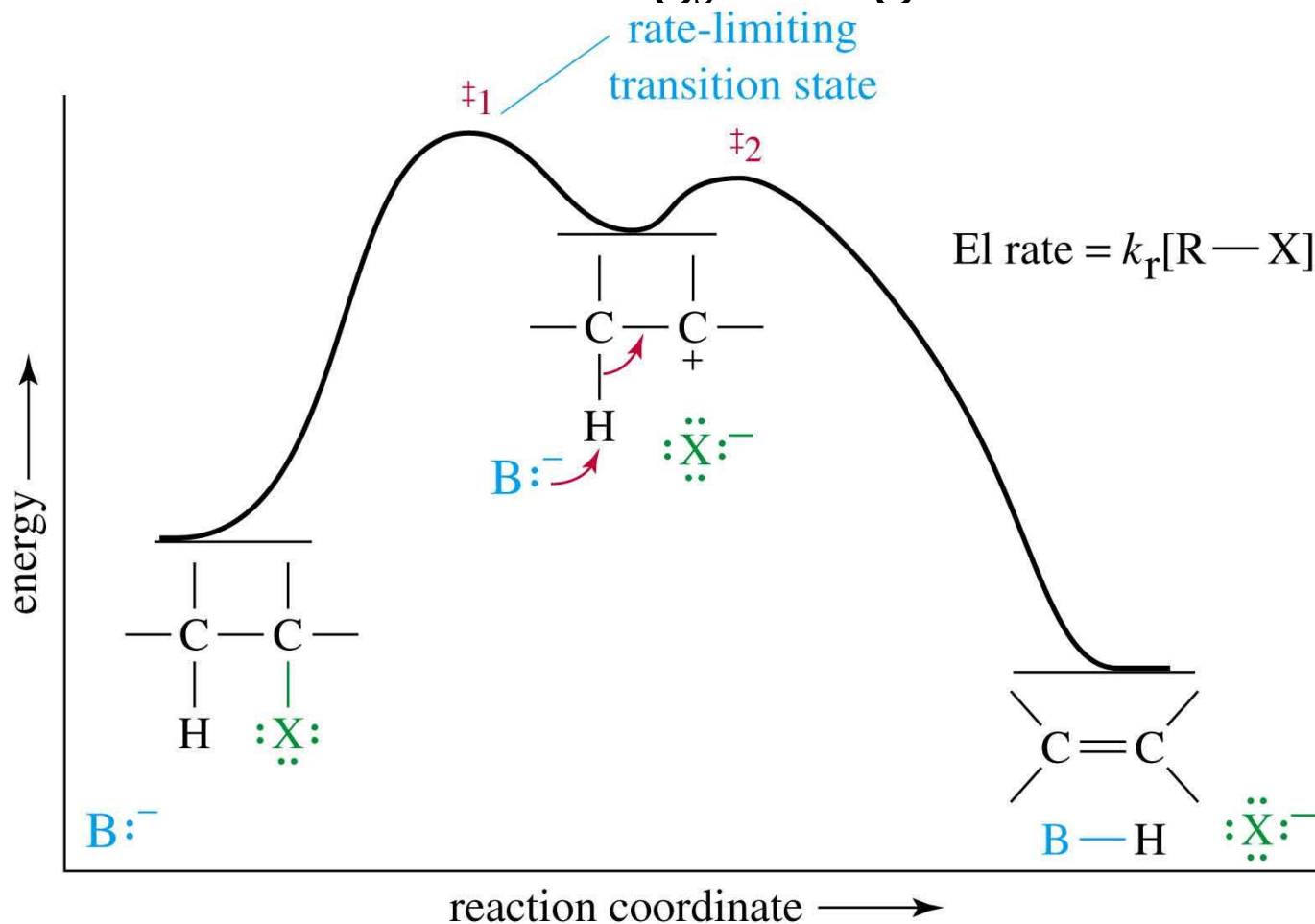


The E1 Reaction





E1 Energy Diagram



- Note: first step is same as S_N1

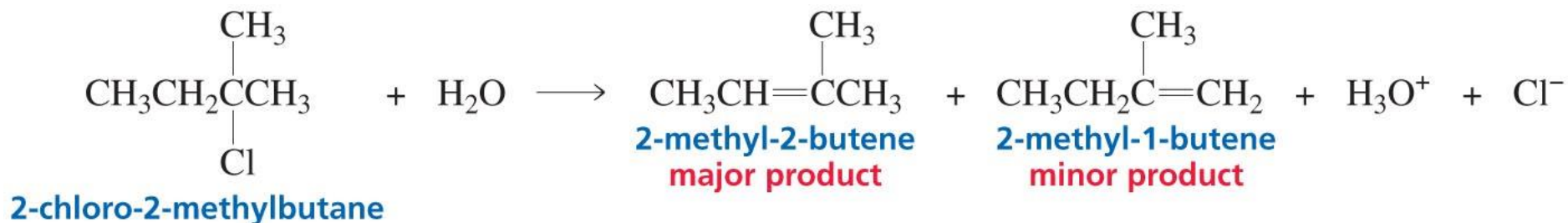
E1 Reaction

- Unimolecular elimination
- Nucleophile acts as base
- Also have S_N1 products (mixture)

$$\text{rate} = k[\text{alkyl halide}]$$

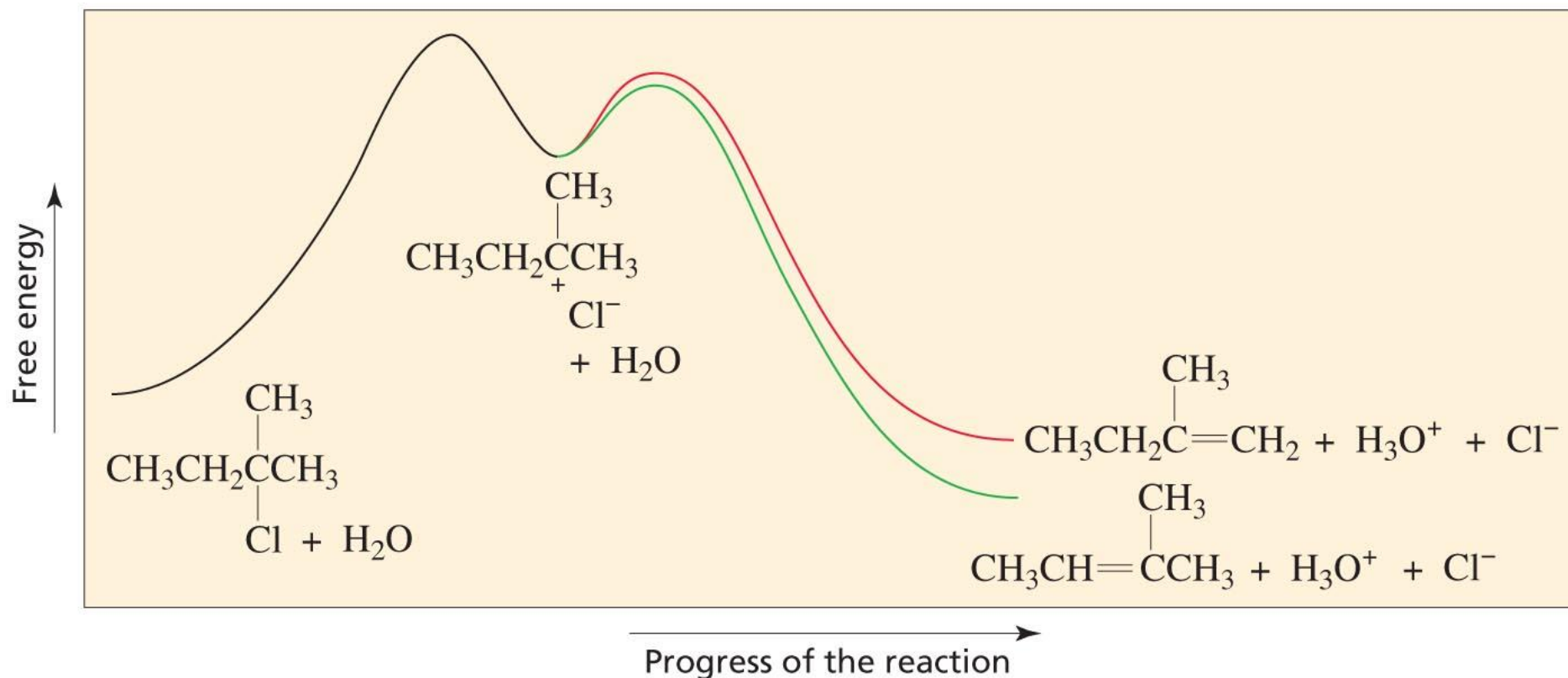
© 2011 Pearson Education, Inc.

The major product in an **E1** reaction is generally the more substituted alkene



© 2011 Pearson Education, Inc.

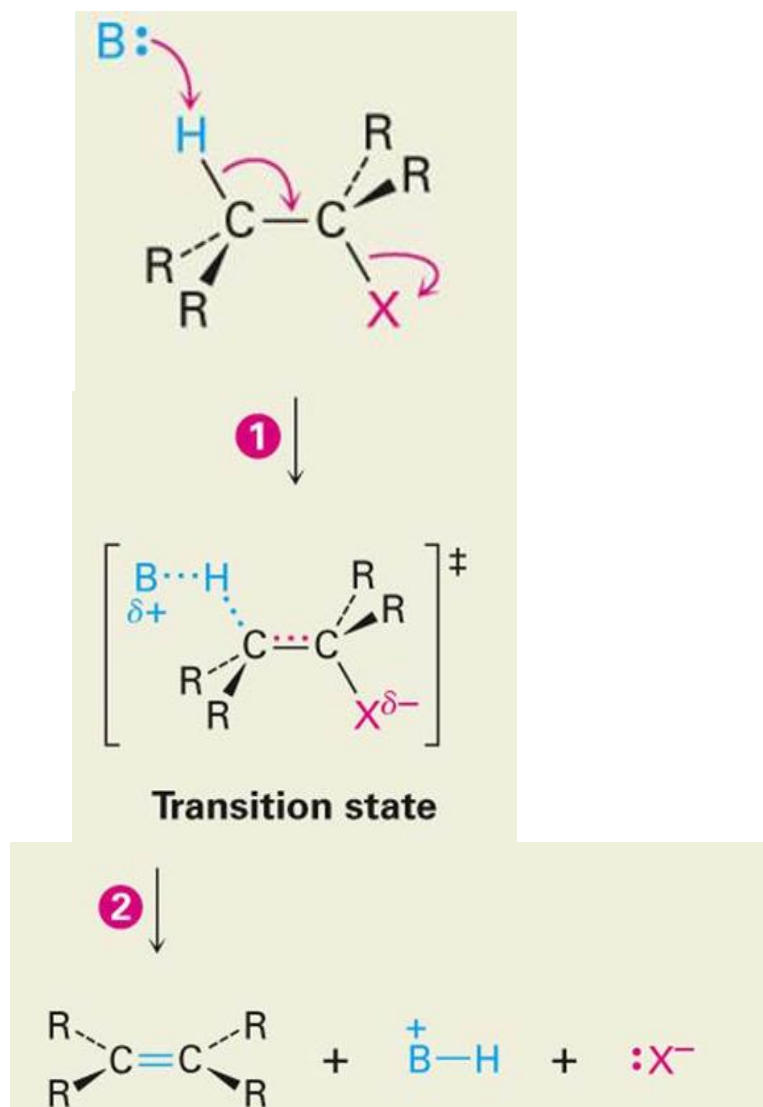
Reaction coordinate diagram for the **E1** reaction of 2-chloro-2-methylbutane



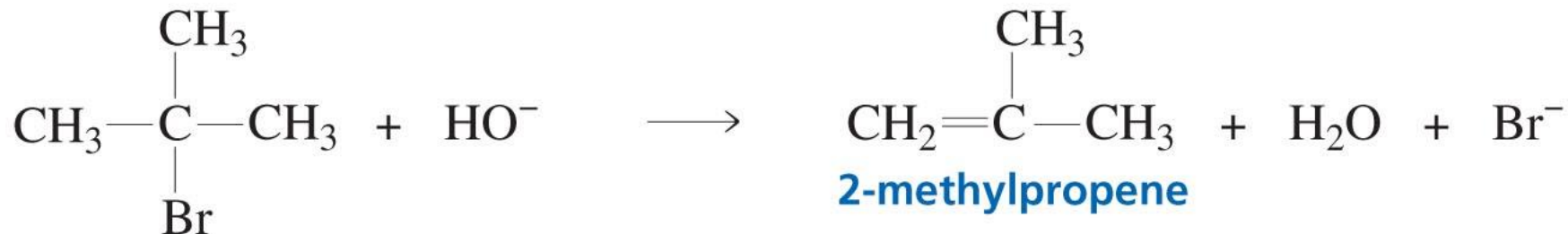
Because the first step is the rate-determining step, the rate of an **E1** reaction depends both on the ease with which the carbocation is formed and how readily the leaving group leaves

The E2 Reaction

- A proton is transferred to base as leaving group begins to depart
- Transition state combines leaving of X and transfer of H
- Product alkene forms stereospecifically
- Rate = $k[\text{RX}][\text{B}]$



The E2 Reaction

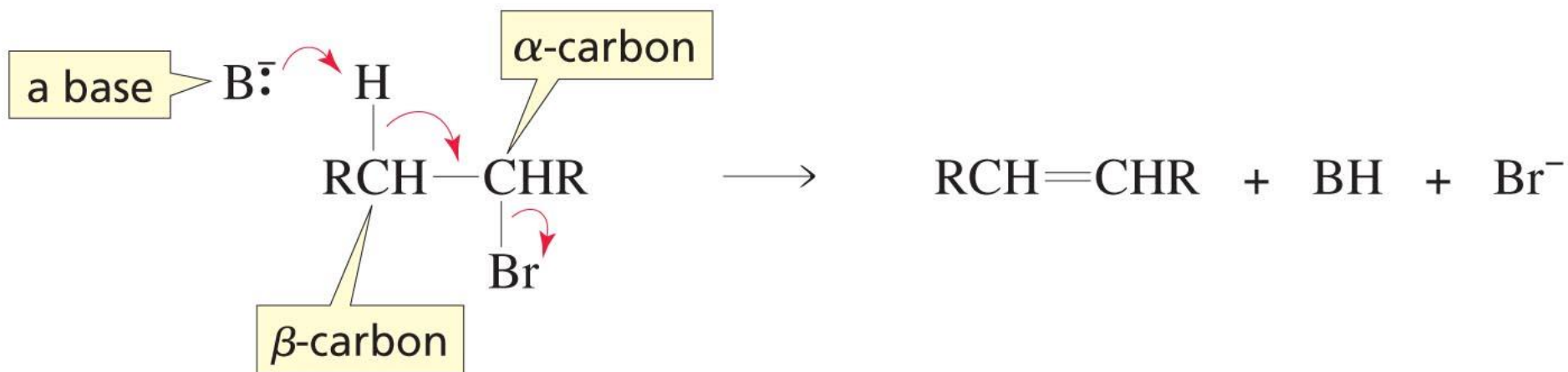


2-bromo-2-methylpropane

© 2011 Pearson Education, Inc.

Hydroxide cannot act as a nucleophile in this reaction because of the bulky tertiary halide. Rather, hydroxide acts as a base and abstracts a proton.

An E2 reaction is also called a β -elimination or a 1,2-elimination reaction:



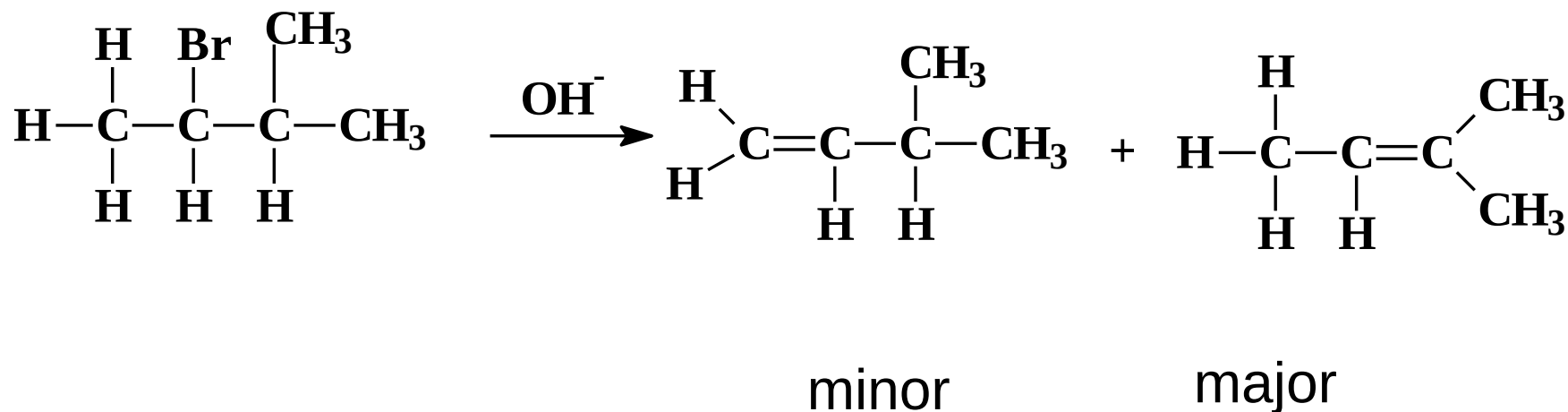
The weaker the base, the better it is as a leaving group

relative reactivities of alkyl halides in an E2 reaction



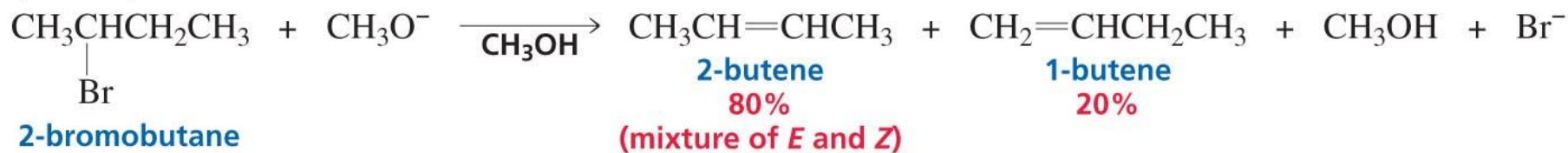
The Regioselectivity of the E2 Reaction

The major product of an E2 reaction is the most stable alkene: (more branch alkene more stable)

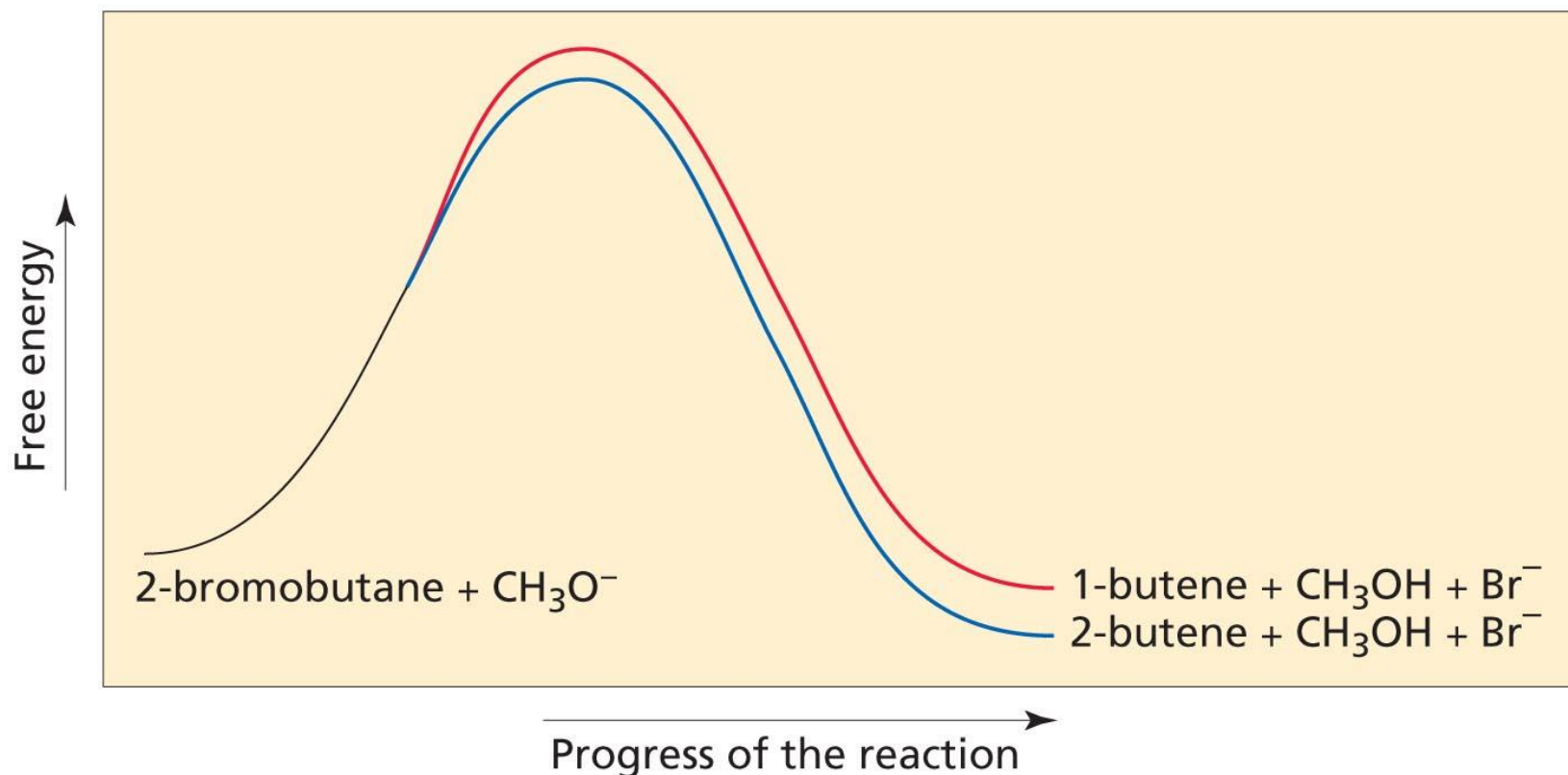


Reaction coordinate diagram for the **E2** reaction of 2-bromobutane and methoxide ion

β -carbons



© 2011 Pearson Education, Inc.



© 2011 Pearson Education, Inc.

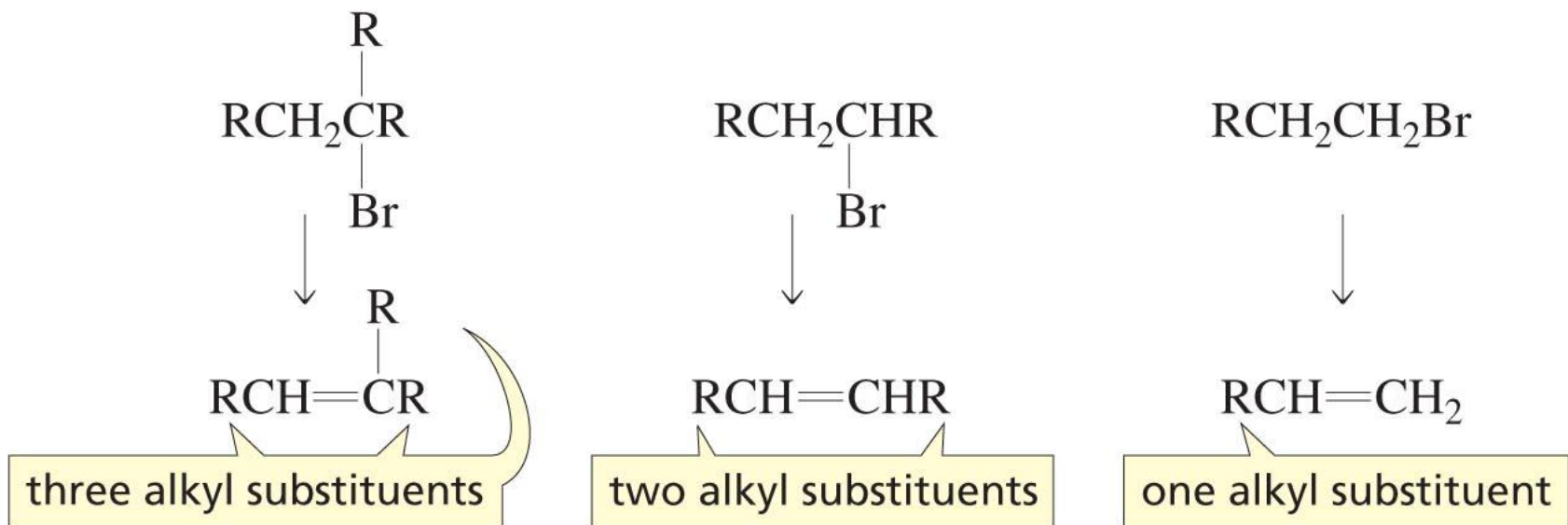
The Zaitsev Rule

The more substituted alkene product is obtained when a proton is removed from the β -carbon that is bonded to the fewest hydrogens

The most stable alkene is generally (but not always) the most substituted alkene

relative reactivities of alkyl halides in an E2 reaction

tertiary alkyl halide > secondary alkyl halide > primary alkyl halide



© 2011 Pearson Education, Inc.

Competition Between E2 and E1 Reactions

Table 9.3 Summary of the Reactivity of Alkyl Halides in Elimination Reactions

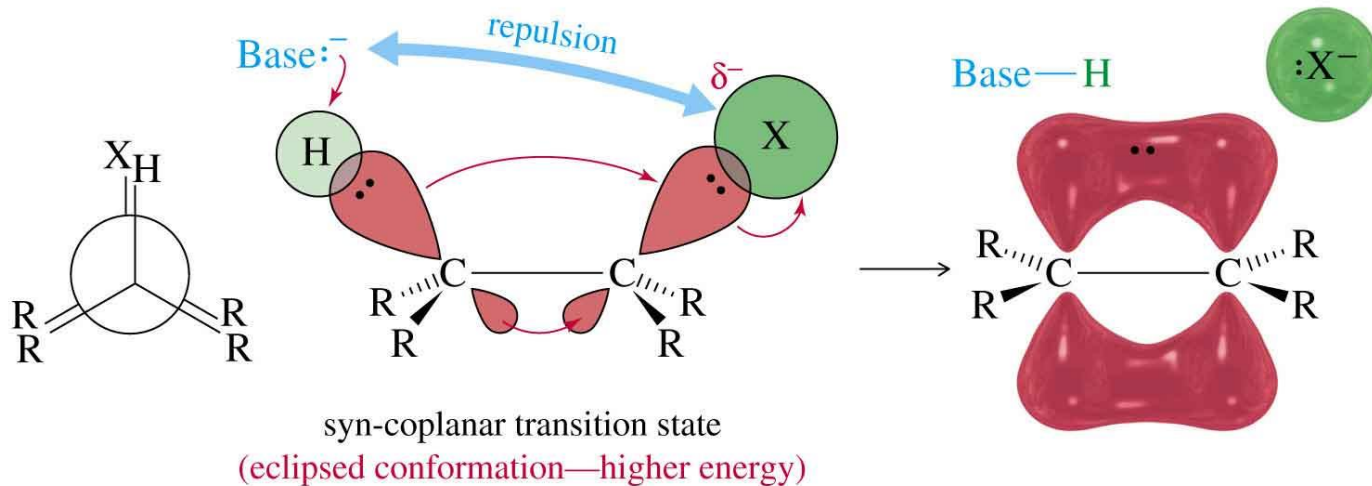
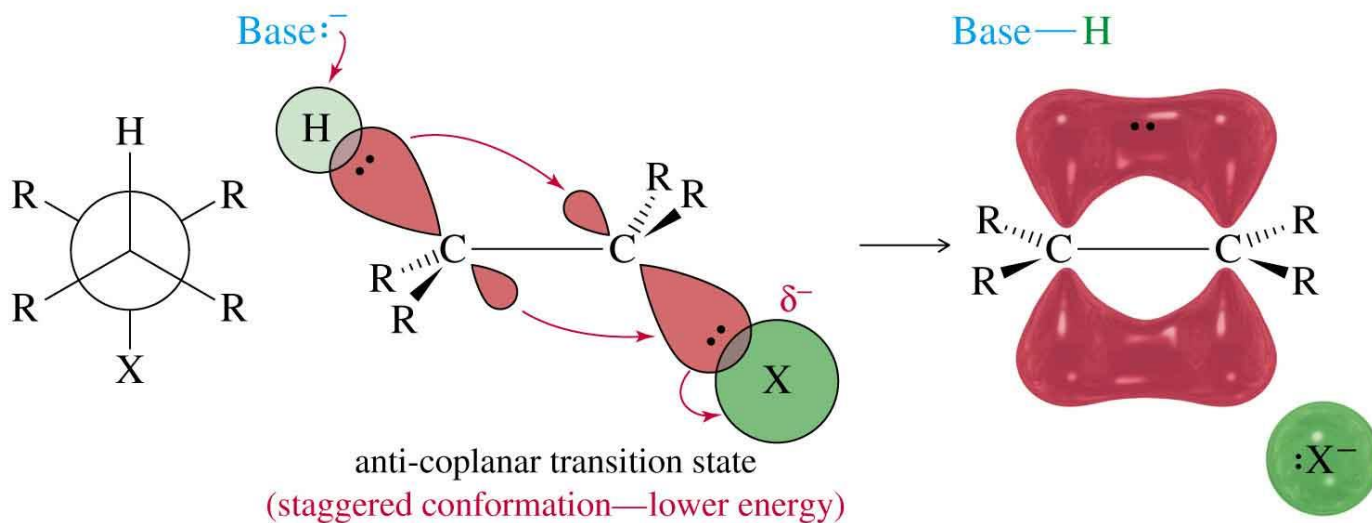
Primary alkyl halide	E2 only
Secondary alkyl halide	E1 and E2
Tertiary alkyl halide	E1 and E2

© 2011 Pearson Education, Inc.

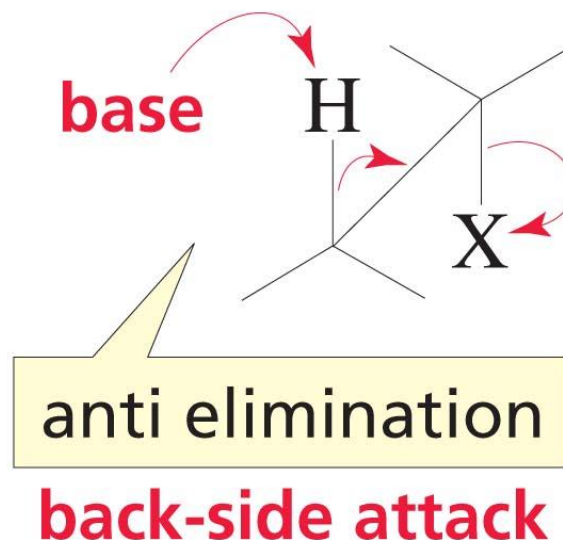
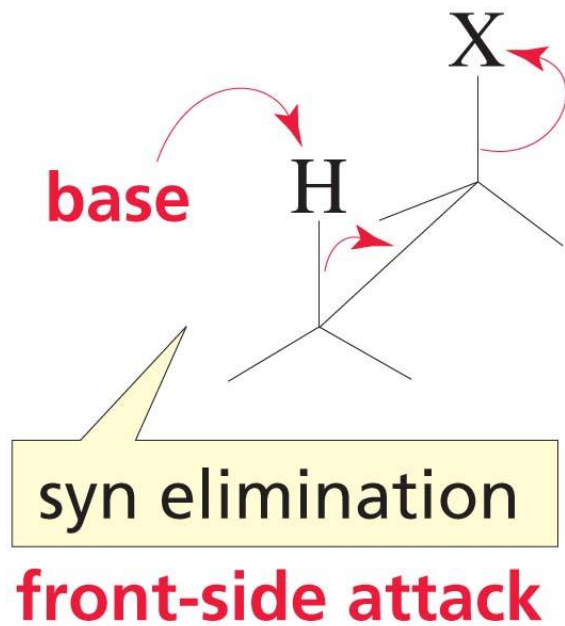
An E2 is favored by a high concentration of strong base and an aprotic polar solvent

An E1 is favored by a weak base and protic polar solvent

E2 Stereochemistry



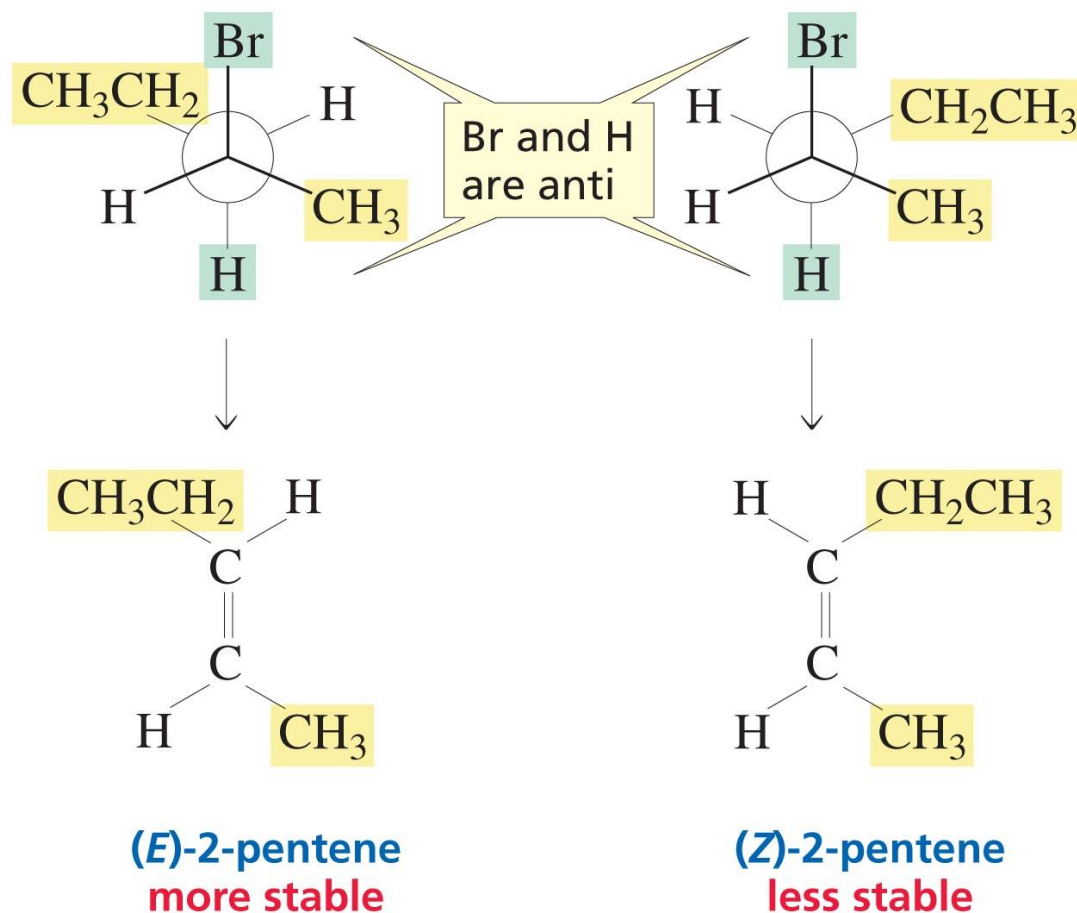
Another view of the E2 reaction mechanism



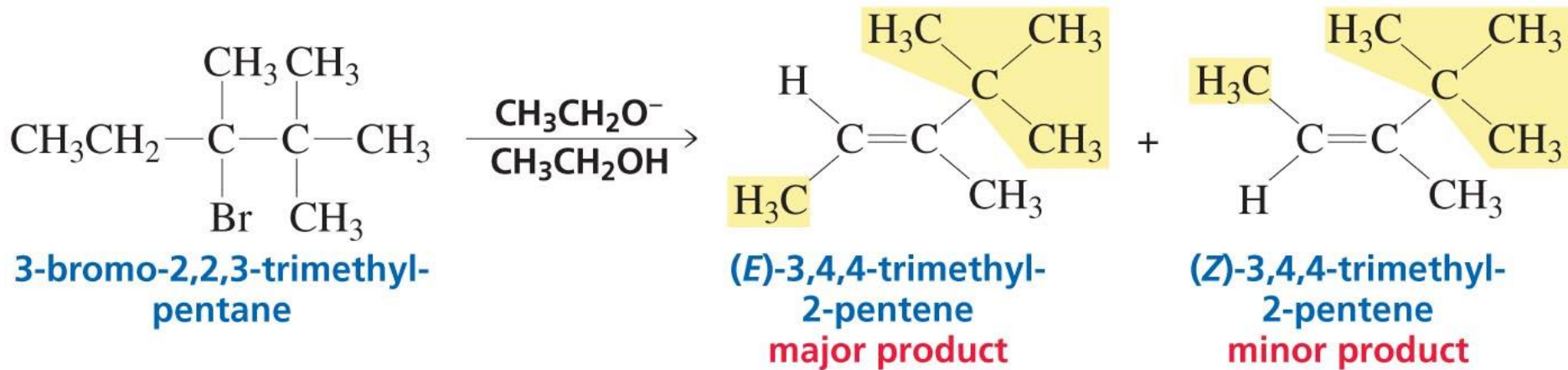
The best overlap of the interacting orbitals is achieved through back-side attack

Anti elimination avoids repulsion of the electron-rich base

Consider the stereoselectivity of the E2 reaction

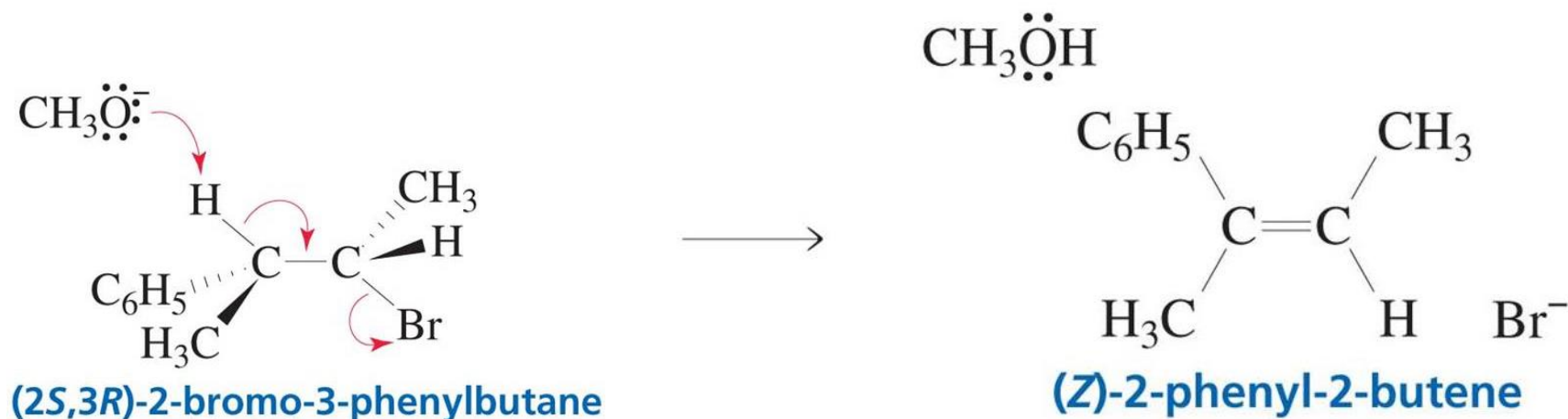
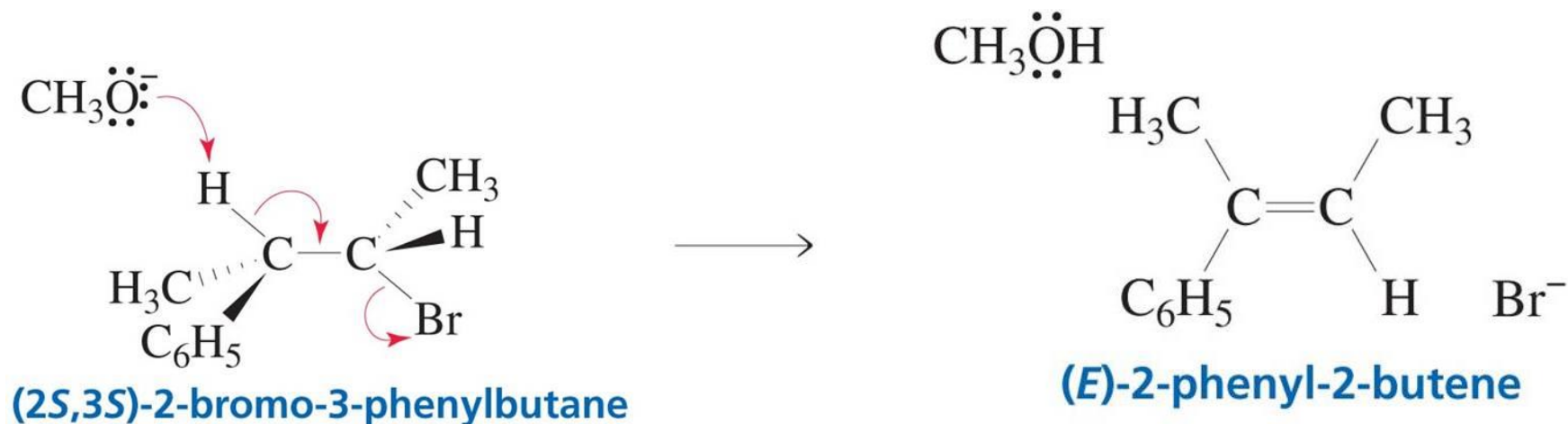


The alkene with the bulkiest groups on opposite sides of the double bond will be formed in greater yield, because it is the more stable alkene



© 2011 Pearson Education, Inc.

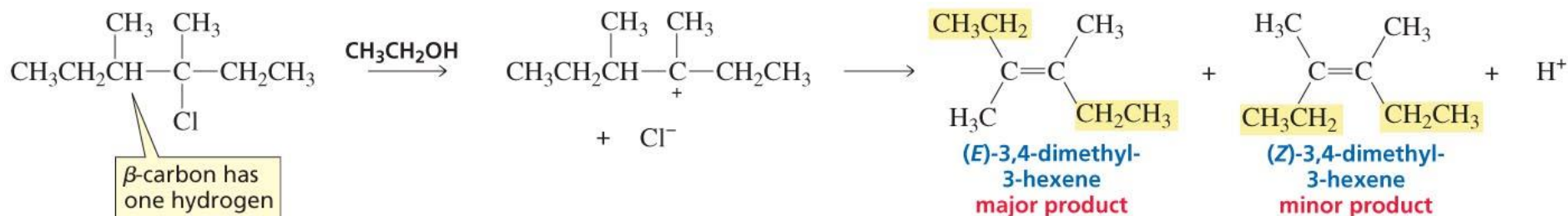
When only one hydrogen is bonded to the β -carbon, the major product (E or Z) from E2 reaction depends on the structure of the alkene



Stereochemistry of the E1 Reaction

The major stereoisomer obtained from an E1 reaction is the alkene in which the bulkiest substituents are on opposite sides of the double bond

Both syn and anti elimination can occur in an E1 reaction

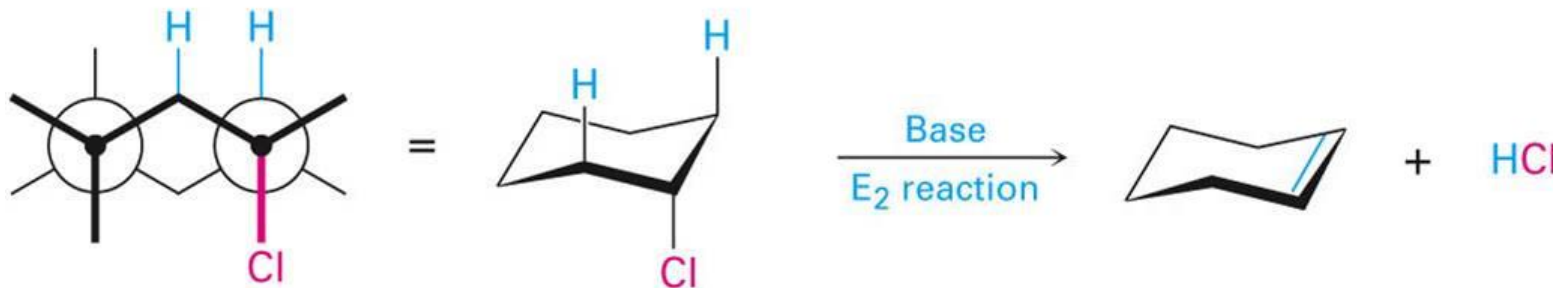


© 2011 Pearson Education, Inc.

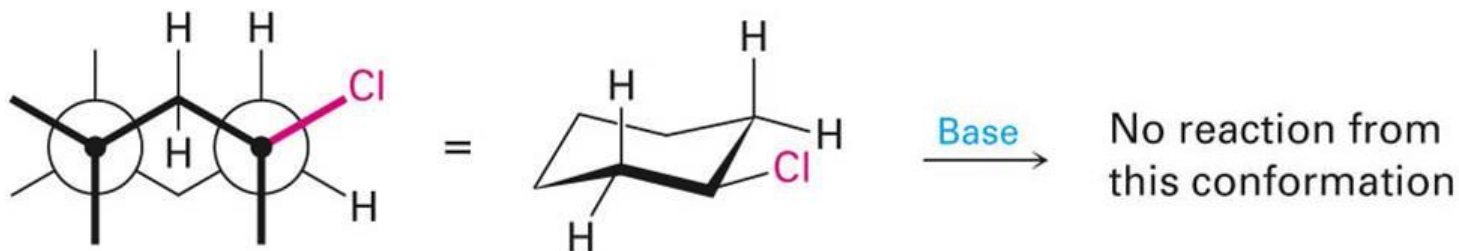
Elimination from Cyclic Compounds

- Abstracted proton and leaving group should align trans-diaxial to be anti periplanar (*app*) in approaching transition state
- Equatorial groups are not in proper alignment

Axial chlorine: H and Cl are anti periplanar

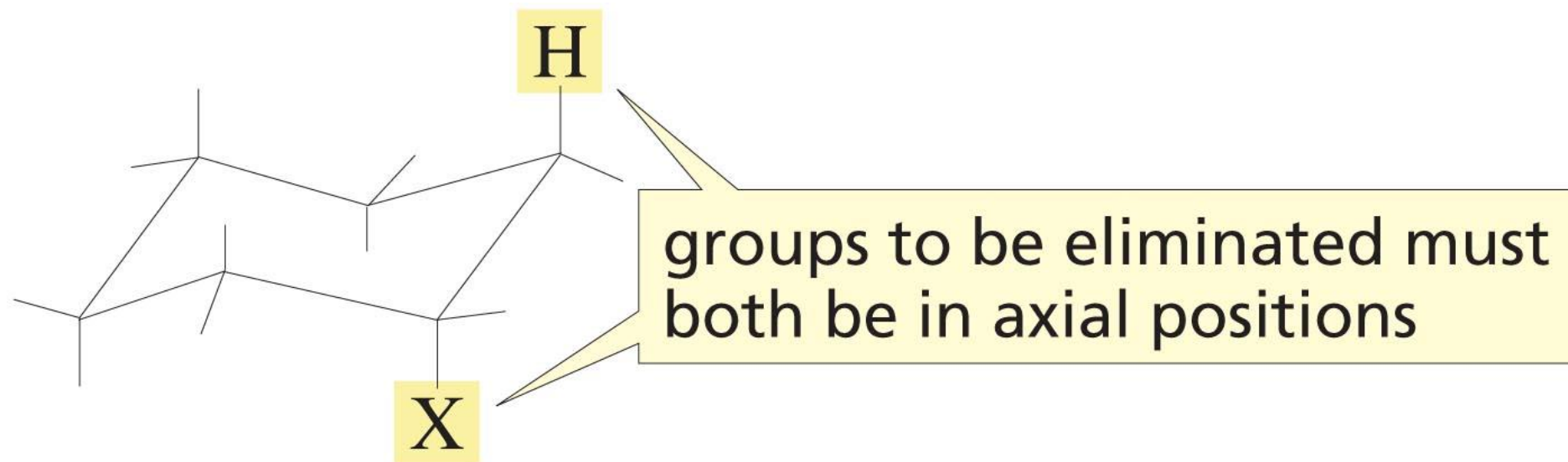


Equatorial chlorine: H and Cl are not anti periplanar



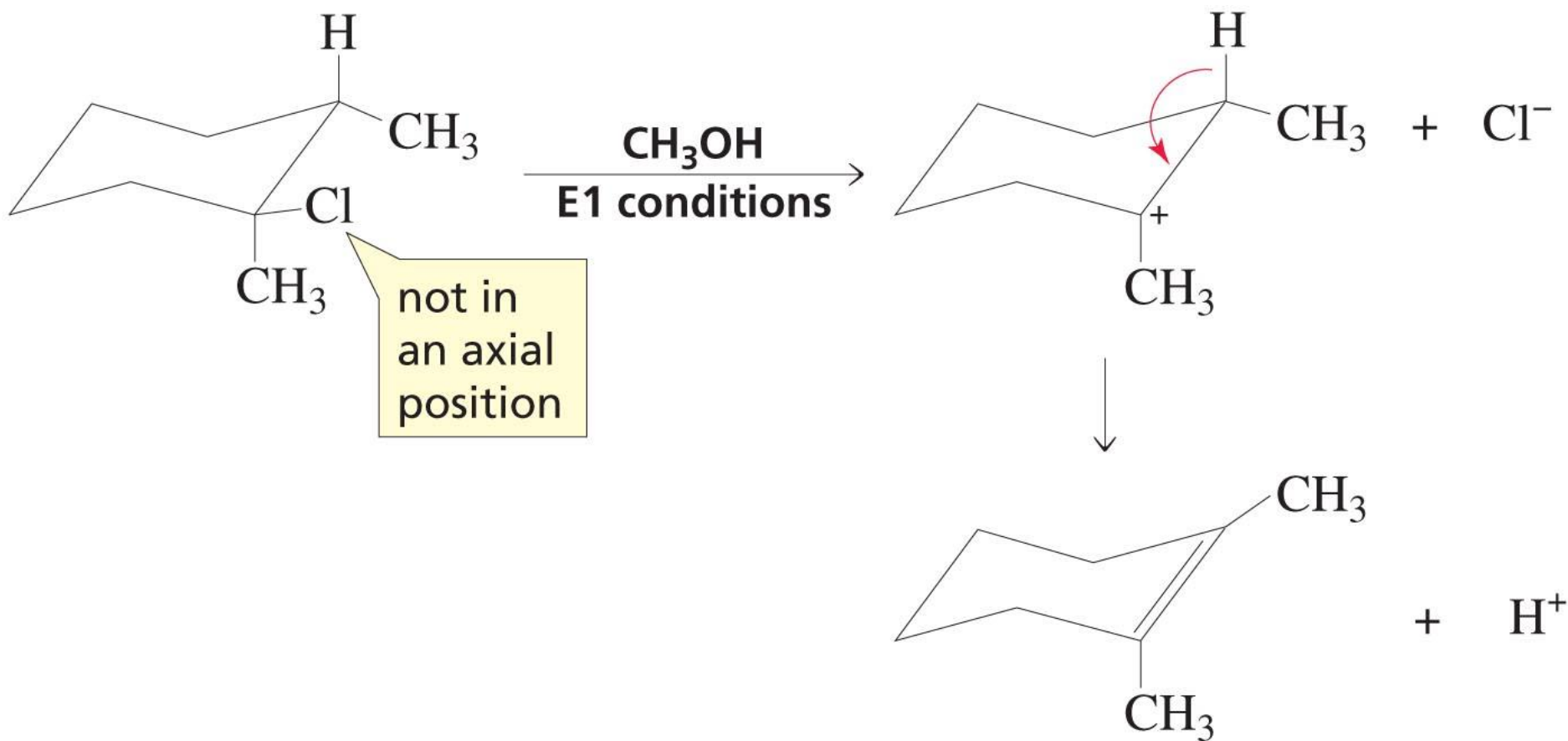
Elimination from Cyclic Compounds

In an E2 reaction, groups to be eliminated must be in axial positions



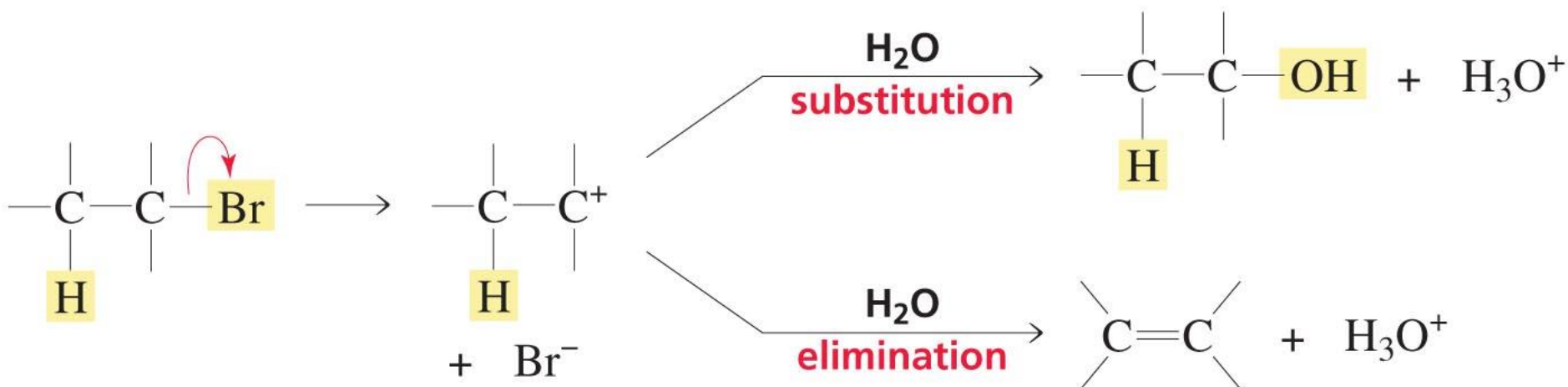
© 2011 Pearson Education, Inc.

E1 Elimination from Cyclic Compounds



No axial orientation required for the E1 reaction.

Consider S_N1/E1 conditions:

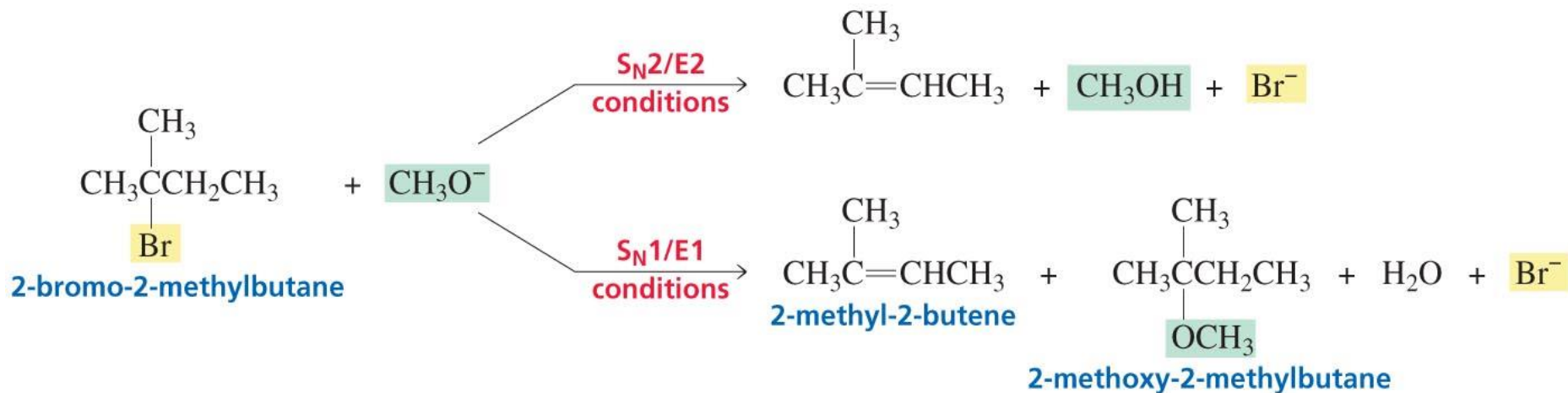


© 2011 Pearson Education, Inc.

The elimination reaction favored at higher temperatures.

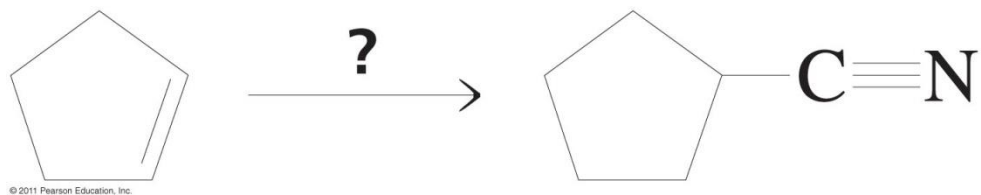
Primary alkyl halides do not form carbocations; therefore they cannot undergo S_N1 and E1 reactions.

The **E2** reaction should be used if you want to synthesize an alkene:

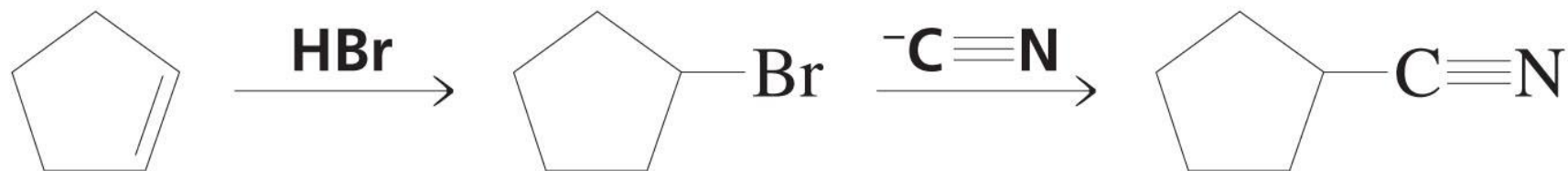


© 2011 Pearson Education, Inc.

Synthesis Example



synthesis



target molecule

© 2011 Pearson Education, Inc.

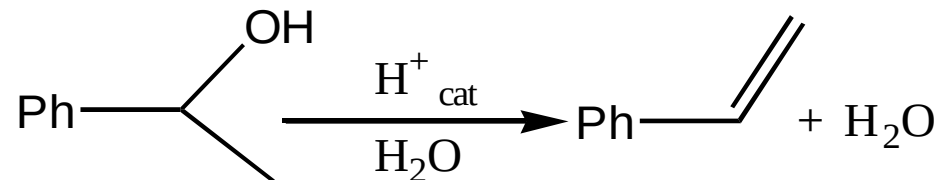
Q: What is the mechanism of the following reaction?

A) SN1

B) SN2

C) E1

D) E2



Q: What is the mechanism of the following reaction?

A) SN1

B) SN2

C) E1

D) E2

